Online supplement to “Neural means and kernel corrections for operator learning”

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Abstract

This supplement accompanies the main text and is available with the code at `https://github.com/yspennstate/neural-means-kernel-corrections` (`paper/supplement.pdf`). Section S1 states the conditional finite-sample bounds for the fitted stages; Section S2 the margin law and the per-coordinate selection results; Section S3 the split-conformal coverage statement; Section S4 the uncontrolled survey of the remaining suite problems and the mirror-symmetry check; Section S5 the effective-dimension identity and its Marčenko–Pastur closed form; Section S6 the additional structural-mechanics results (ablation, error localization, spectra, uncertainty quantification, cost); Section S7 the OCO-2 input-metric study and the data-scaling and ClimSim results; Sections S8–S10 all proofs, the implementation and compute record, and the numerical checks of every stated identity and inequality; and Section S11 the paired centering, correction-label, and kernel-grid sensitivity experiments and reconstruction from saved fields. Numbering of the form Sn refers to this document; all other references point to the main text.

S1 Guarantees for the fitted stages

S1.1 Full-map symmetry averaging does not increase the risk

The elasticity problem behind the benchmark is invariant under the reflection $x_1 \mapsto 1 - x_1$: the domain, the boundary partition, and the constitutive model are all symmetric, the input distribution has a reflection-invariant covariance, and reflecting the load reflects the stress field. On the discrete grid this is expressed by two permutation matrices: $S \in \mathbb{R}^{p\times p}$ reverses the load samples and $T \in \mathbb{R}^{q\times q}$ reverses the rows of the stress field. Both are orthogonal involutions. Section 2 describes a direct data-driven check of the equivariance $G(Su) = T\,G(u)$; we treat it as an assumption here.

Proposition S1.1 (symmetrization)**.** *Assume $G(Su) = TG(u)$ for μ-a.e. u, that μ is invariant under S, and that T is an orthogonal involution ($T^2 = I$). For a measurable $\widehat{G} : \mathcal{U} \to \mathcal{V}$ define its symmetrization*

$$\widehat{G}_S(u) \;=\; \tfrac{1}{2}\Big(\widehat{G}(u) + T\,\widehat{G}(Su)\Big).$$

Then, for every loss $\ell(\cdot, v)$ that is convex in its first argument and satisfies $\ell(Tw, Tv) = \ell(w, v)$,

$$\mathbb{E}_{u\sim\mu}\,\ell\big(\widehat{G}_S(u), G(u)\big) \;\le\; \mathbb{E}_{u\sim\mu}\,\ell\big(\widehat{G}(u), G(u)\big).$$

Proof. See Supplement S8.1.

The relative error satisfies both hypotheses when the target norm is nonzero. The proposition guarantees that test-time reflection averaging cannot increase population risk under exact equivariance and distributional invariance. For a finite training sample, Jensen's inequality also shows that the reflection-augmented empirical loss upper-bounds the empirical loss of the symmetrized predictor. The objectives are generally different, so the proposition does not guarantee that training with augmentation improves the fitted model. The near-mirror checks support exact symmetry of the distributed solver output. In the campaign, the averages improve twenty-nine of thirty members across the two MLP variants and the refiner, and worsen one by a thousandth of a point. The refiner's operation requires the distinction below. The convexity argument is standard; Lyle et al. (2020) give a general account.

A supplied conditioning field is an additional input. The proposition covers the full-map average f_S. A sufficient condition for the implemented supplied-field average to equal f_S is $h(Su) = Th(u)$; the fitted kernel is not constrained to satisfy this condition. For a fitted refiner $F(u, h)$ and a fixed kernel predictor $h(u)$, define $f(u) = F(u, h(u))$. The proposition concerns

$$f_S(u) = \tfrac{1}{2}\big(F(u, h(u)) + TF(Su, h(Su))\big).$$

The implementation instead uses $Th(u)$ as the second branch's conditioning field. Thus $h(Su) = Th(u)$ is a sufficient additional condition for identifying the implemented average with f_S. Equivariance of the target and invariance of the load distribution do not by themselves give this property to a fitted kernel predictor, and training the refiner with randomly reflected triples does not force exact equivariance; the non-increase guarantee for its averaging operation is conditional on that identity.

There is a quantitative way to state the missing requirement. Write $b(u) = \frac{1}{2}(F(u, h(u)) + TF(Su, Th(u)))$ for the implemented average and $R(g) = \mathbb{E}\|g(u) - G(u)\|_2/\|G(u)\|_2$, assuming nonzero targets almost surely and finite quantities below. Under the proposition's exact symmetry assumptions,

$$\begin{aligned} R(b) &\le R(f_S) + \frac{1}{2}\mathbb{E}\frac{\|F(Su, Th(u)) - F(Su, h(Su))\|_2}{\|G(u)\|_2} \\ &\le R(f) + \frac{1}{2}\mathbb{E}\frac{\|F(Su, Th(u)) - F(Su, h(Su))\|_2}{\|G(u)\|_2}. \end{aligned} \tag{S1.1}$$

Indeed, the triangle inequality bounds the risk difference by $\mathbb{E}\|b - f_S\|_2/\|G\|_2$, and orthogonality of T gives the displayed defect exactly; the second inequality is Proposition S1.1. If $F(u, \cdot)$ is uniformly L_h-Lipschitz, the additional term is at most $(L_h/2)\mathbb{E}[\|Th(u) - h(Su)\|_2/\|G(u)\|_2]$. This defect and the Lipschitz constant are not measured in the experiments, so the inequality is a structural statement rather than a numerical bound.

The distinction can matter even with an exactly symmetric target. Let the input take the values $+1$ and -1 with equal probability, let S exchange them, and let T swap two output coordinates. Set $G(u) = (1, 1)$, $h(+1) = (1, 1)$ and $h(-1) = (3, 3)$. Define $F(+1, z) = z$ and $F(-1, z) = (4, 4) - z$. The four evaluations are

$$\begin{aligned} &F(+1, h(+1)) = (1, 1), \quad F(-1, Th(+1)) = (3, 3), \\ &F(-1, h(-1)) = (1, 1), \quad F(+1, Th(-1)) = (3, 3). \end{aligned}$$

Thus $f(u) = G(u)$ at both inputs, and the full-map average also has zero error. Averaging with $Th(u)$ instead gives $(2, 2)$ at both inputs, with relative error one. This finite example disproves a general guarantee for the supplied-field operation and attains equality in (S1.1) (here $L_h = 1$). It is not a model of the measured benchmark. The accompanying `campaign/refiner_symmetry_diagnostic.py` checks the construction and independently recomputes the ten recorded refiner differences.

S1.2 What the kernel-flow regularizer estimates

The kernel-flow (KF) loss of Owhadi and Yoo (Owhadi and Yoo, 2019), in the ℓ^2 variant used by Yoo and Owhadi (2021) to regularize deep networks, motivates the optional regularizer in our ablation study. We first state the identity for exact interpolation, then describe the implementation's regularization and differentiation choices. Let $z_i = \phi_\theta(u_i)$ be the features of a batch $B = \{1, \dots, b\}$ (b even), let k be a positive definite kernel on feature space, and for $C \subset B$, $|C| = b/2$, let I_C denote the k-interpolant of $\{(z_j, v_j) : j \in C\}$. The batch KF loss is

$$e_2(\theta; C) \;=\; \sum_{i \in B} \big\| v_i - I_C(z_i) \big\|_2^2. \tag{S1.2}$$

Proposition S1.2 (KF loss is a leave-half-out estimate)**.** *Let C be drawn uniformly among the subsets of B of size $b/2$ and assume k restricted to $\{z_i\}_{i \in B}$ is strictly positive definite. Then $e_2(\theta; C) = \sum_{i \in B \setminus C} \|v_i - I_C(z_i)\|_2^2$, and*

$$\mathbb{E}_C\big[e_2(\theta; C)\big] \;=\; \frac{b}{2}\, \mathbb{E}_{C,\, i \sim \mathrm{Unif}(B \setminus C)}\big[\big\|v_i - I_C(z_i)\big\|_2^2\big],$$

i.e. $\frac{2}{b}\, e_2$ is an unbiased estimator, over the split randomness, of the expected squared error committed on a held-out point by the kernel predictor trained on a random half of the batch. When (B, C) are resampled at every step, the stochastic gradient $\nabla_\theta e_2(\theta; C)$ is unbiased for $\nabla_\theta\, \mathbb{E}_{B,C}[e_2(\theta; C)]$ whenever the expectation and derivative commute (e.g. under local Lipschitz domination).

Proof. See Supplement S8.2.

The implemented ablation uses normalized targets and the mean squared loss over all bq batch-output entries, with predictor $K_{BC}(K_{CC} + 10^{-3} I)^{-1} Y_C$. Its errors on C need not vanish: on those rows the residual is $10^{-3}(K_{CC} + 10^{-3} I)^{-1} Y_C$. Consequently the exact leave-half-out identity above does not apply to the regularized loss. The learned Gaussian bandwidth multiplier is combined with a batch median distance computed under `no_grad`; the resulting gradient also omits the dependence of that median on the features. It is a stop-gradient training rule, not the full derivative in Proposition S1.2. These choices are fixed in the reported ablation. Neither the ideal identity nor its regularized analogue estimates the population risk of the adaptively trained network.

S1.3 Capacity of stacking with fixed members

Convexity bounds the loss of a convex combination by the corresponding weighted average of member losses. A separate uniform-deviation argument bounds the cost of fitting the weights when the candidate members are fixed independently of an i.i.d. validation sample and the losses have a population bound. These conditions are part of the following result; they are not assumed for the pipeline's validation split.

Proposition S1.3 (validation stacking). *Let $f_1, \ldots, f_M : \mathcal{U} \to \mathcal{V}$ be fixed maps, let $\Delta = \{w \in \mathbb{R}^M : w_m \geq 0, \sum_m w_m = 1\}$, and for $w \in \Delta$ write $F_w = \sum_m w_m f_m$. Assume the members' per-sample relative errors are bounded: $\ell(f_m(u), G(u)) \leq B$ for μ-a.e. u and every m.*

(i) For every $w \in \Delta$, $\ell(F_w(u), G(u)) \leq \sum_m w_m \, \ell(f_m(u), G(u))$ pointwise; in particular $R(F_w) \leq \sum_m w_m R(f_m) \leq \max_m R(f_m)$, and $\ell(F_w(u), G(u)) \leq B$.

(ii) Let $\widehat{w}$ minimize the empirical risk $\widehat{R}(w) = \frac{1}{m} \sum_{i=1}^{m} \ell(F_w(\tilde{u}_i), G(\tilde{u}_i))$ over Δ, computed on an i.i.d. validation sample $\tilde{u}_1, \ldots, \tilde{u}_m \sim \mu$ independent of $f_1, \ldots, f_M$. Then for every $\delta \in (0,1)$, with probability at least $1 - \delta$,

$$R(F_{\widehat{w}}) \;\leq\; \min_{w \in \Delta} R(F_w) \;+\; \frac{8B}{m} \;+\; B\sqrt{\frac{2\big((M-1)\log\big(m(M-1)+1\big) + \log(2/\delta)\big)}{m}}.$$

Proof. See Supplement S8.3.

With $M = 5$, $m = 1000$ and $\delta = 0.05$, the excess term is about $0.28B$. Substituting $B = 0.25$, the largest observed member error, gives roughly seven percentage points, conditional on that value being a genuine uniform population bound. Even that conditional value is too large to make the selection error negligible. The result supplies the scaling $\sqrt{M \log m / m}$; the much smaller validation-to-test gaps are empirical measurements.

The stack the pipeline deploys on the structural-mechanics benchmark is finer than a single convex vector: the weights are affine and vary per grid point, fitted by ridge on the validation split, and the choice between this object and the global convex stack is itself made on held-out data (fit both on one half of the split, keep whichever wins on the other half, refit the winner on the full split). Proposition S1.3 does not cover that construction. The following two statements give conditional capacity bounds for related fixed-candidate selection and affine classes.

Lemma S1.4 (validated selection). *Let $g_1, \ldots, g_K$ be fixed maps from $\mathcal{U}$ to $\mathcal{V}$. Assume $\ell(g_k(u), G(u)) \leq B$ for μ-a.e. u and every k, and let $\widehat{k}$ minimize the empirical risk over an i.i.d. sample of size m' drawn from μ independently of everything used to build the g_k. Then, with probability at least $1 - \delta$,*

$$R(g_{\widehat{k}}) \;\leq\; \min_{k \leq K} R(g_k) \;+\; B\sqrt{\frac{2\log(2K/\delta)}{m'}}.$$

Proof. See Supplement S8.4.

The half-split comparison uses $K = 2$ and $m' = 500$. The lemma applies only if both candidate stacks were constructed independently of the judging half. Refitting a winner changes the fitted map; the next proposition bounds affine fitting under its own independence and boundedness assumptions.

Proposition S1.5 (per-pixel affine stacking)**.** *Fix members $f_1, \dots, f_M$, write $D = \sqrt{M+1}$, and for each output coordinate j let $\varphi_j(u) = (f_1(u)_j, \dots, f_M(u)_j, b) \in \mathbb{R}^{M+1}$, where b bounds all member and target values: $|G(u)_j| \le b$, $|f_m(u)_j| \le b$ for μ-a.e. u and all j, m. Consider the per-coordinate affine class $\{u \mapsto w^\top \varphi_j(u) : \|w\|_2 \le W\}$ and the coordinate risks $R_j(w) = \mathbb{E}\,(w^\top \varphi_j(u) - G(u)_j)^2$, estimated on an i.i.d. validation sample of size m independent of the members.*

(i) If $\widehat{w}_j$ minimizes the empirical risk of coordinate j over the class, then with probability at least $1-\delta$,

$$\frac{1}{q}\sum_{j=1}^{q}\Big[R_j(\widehat{w}_j) - \min_{\|w\|_2 \le W} R_j(w)\Big]$$
$$\le \frac{8\,Wb^2 D\,(1+WD) + 2\,b^2(1+WD)^2\sqrt{\frac{1}{2}\log(4q/\delta)}}{\sqrt{m}}.$$

(ii) If instead $\widehat{w}_j$ minimizes the ridge objective $\widehat{R}_j(w) + \lambda\|w\|_2^2$ with $\lambda \ge 0$ and satisfies $\|\widehat{w}_j\|_2 \le W$, the same bound holds with an additional λW^2 on the right.

Proof. See Supplement S8.5.

For fixed M, b, W, this is an $m^{-1/2}$ bound on coordinatewise mean-squared risk, with logarithmic dependence on the number of coordinates. It is not a relative-error bound. Its constants depend polynomially on b and W; substituting the measured values makes the bound exceed the realized excess by many orders of magnitude. Those substitutions are illustrative, since a finite observed maximum is not a population bound.

A fixed grid of residual corrections gives a further composite class. For each configuration, correction of training residuals is affine in the stack weights, so the same uniform-deviation argument applies with a transformed feature map. The following result requires the training arrays, members and correction configurations to be fixed independently of validation; it does not assert a guarantee for the deployed sequence of checkpoint selection, stacking, subsample tuning and refitting.

Corollary S1.6 (capacity bound for a fixed composite correction class)**.** *Assume the setting of Proposition S1.5, and let Γ be a finite set of correction configurations, each defining weights $c_\gamma(u) \in \mathbb{R}^{n_{\mathrm{tr}}}$ with the corrected predictor*

$$h_{w,\gamma}(u)_j \;=\; w_j^\top \varphi_j(u) + c_\gamma(u)^\top\big(Y_j - \Phi_j w_j\big),$$

where $Y_j \in \mathbb{R}^{n_{\mathrm{tr}}}$ collects the training targets of coordinate j and Φ_j the members' training predictions; $\gamma = 0$ (no correction, $c_0 \equiv 0$) is included in Γ. Suppose the correction weights are uniformly bounded, $\sup_{u,\gamma}\|c_\gamma(u)\|_1 \le \kappa$. Assume that Y_j, Φ_j, the members and all c_γ are fixed independently of the validation sample, and that their training target and feature entries satisfy the same absolute bound b. For any validation-measurable choice $(\widehat{w}, \widehat{\gamma})$ with $\|\widehat{w}_j\|_2 \le W$ and $\widehat{\gamma}$ minimizing the empirical validation risk, aggregated over coordinates, among $\gamma \in \Gamma$ at the fitted $\widehat{w}$ (the grid choice is shared across coordinates), with probability at *least* $1-\delta$,

$$\frac{1}{q}\sum_j R_j\big(h_{\widehat{w},\widehat{\gamma}}\big) \le \min_{\gamma\in\Gamma}\frac{1}{q}\sum_j R_j\big(h_{\widehat{w},\gamma}\big) + \frac{2(1+\kappa)^2 b^2(1+WD)}{\sqrt{m}}\left(4WD + (1+WD)\sqrt{\frac{\log(4q|\Gamma|/\delta)}{2}}\right).$$

Since $\gamma = 0$ is among the candidates, the oracle on the right is at most the fitted stack's own risk, and combining with Proposition S1.5(ii) bounds the composite risk against the best affine stack, provided the hypotheses of both results hold (with failure probabilities allocated between them). Here b and W must be population and class bounds; the observed values are diagnostics of them. For kernel ridge weights, κ is supplied by the next lemma.

Proof. See Supplement S8.7.

The hypothesis on the correction weights is not an empirical assumption. In the pipeline each $\gamma = (s, \lambda)$ is a length scale and a nugget, and the weights are those of kernel ridge regression, $c_\gamma(u) = (K_\gamma + n\lambda I)^{-1}k_\gamma(X, u)$, so that $\widehat{r}(u) = c_\gamma(u)^\top R$ in the notation of Section 3.3. Their ℓ^1 norm is bounded uniformly in u by the nugget alone.

Lemma S1.7 (uniform bound on kernel ridge weights)**.** *Let k be a positive definite kernel with $k(u,u) \le k_{\max}$ for all u, let $X = (u_1, \ldots, u_n)$ be any design with Gram matrix K, and let $\lambda > 0$. The kernel ridge weights $c_\lambda(u) = (K + n\lambda I)^{-1}k(X, u)$ satisfy, for every u,*

$$\|c_\lambda(u)\|_1 \;\le\; \sqrt{n}\,\|c_\lambda(u)\|_2 \;\le\; \frac{1}{2}\sqrt{\frac{k(u,u)}{\lambda}} \;\le\; \frac{1}{2}\sqrt{\frac{k_{\max}}{\lambda}},$$

and the constant $\frac{1}{2}$ cannot be improved: for every $\lambda \in (0,1)$ there are a kernel, a two-point design and a point u at which the first and last members are equal. In particular, for the residual correction of Section 3.3 — a Matérn kernel with $k(u,u) = 1$ and the nugget grid $\lambda \in \{10^{-6}, 10^{-5}, 10^{-3}\}$ — the hypothesis of Corollary S1.6 holds, whatever the length scale and the design, with

$$\kappa_{\mathrm{an}} \;:=\; \tfrac{1}{2}\max_{\gamma\in\Gamma}\lambda_\gamma^{-1/2} \;=\; 500.$$

Proof. See Supplement S8.8.

The proof is a few lines in Section S8: the Gram matrix of the design together with u is positive semidefinite, which confines $k(X,u)$ to the range of K with $k(X,u)^\top K^+ k(X,u) \le k(u,u)$, and $\mu/(\mu + n\lambda)^2 \le 1/(4n\lambda)$ for every eigenvalue μ. The bound is the right kind of constant for the corollary: it depends on nothing that is estimated. It is also why the corollary is a capacity bound and not a certificate. The excess term carries $(1+\kappa)^2$, so at $\kappa_{\mathrm{an}} = 500$ the corollary's right-hand side is, like that of Proposition S1.5 before it, vacuous at the stated constants by many orders of magnitude. The realized scale of the weights is far smaller: on a test probe of five hundred points the largest ℓ^1 norm over the grid averages 32.2 over the ten seeds (Section S10), about a fifteenth of κ_{an}, so that evaluating the display at the probe value would shrink it by a factor of about two hundred and forty. But a supremum over a finite probe is a lower bound on the constant the corollary needs, not the constant. What the corollary establishes is conditional: for candidates fixed independently of the validation sample, in the coordinatewise mean-squared metric, an oracle inequality of the ordinary $m^{-1/2}$ form holds with constants that are assumed as population bounds or, for κ, given in closed form. It is not a bound on the estimator the pipeline fits: that estimator is selected by mean relative L^2 error rather than by squared coordinate risk, its kernel stages are tuned on an 8000-row subsample and the winner is refit on all 19000 rows, and the same validation split selects checkpoints and members — none of which the statement covers. Two cautions apply to every statement in this section and are repeated in the main text: the candidates are assumed fixed independently of the validation sample, which the deployed pipeline does not satisfy, since the same split selects checkpoints, members and, on OCO-2, coordinates; and the constants B, b, L, W are population bounds, for which the text substitutes observed maxima, so the evaluated values are conditional capacity calculations and not guarantees for the fitted estimator. The split-conformal result of Section S3 gives marginal coverage when its separate calibration assumptions hold; Theorem 6.9 is a conditional worst-case bound whose constant is not observed.

S2 Decision rules: margins and per-coordinate selection

The two-member criterion is exact for population second moments. Applying it to estimates introduces a separate estimation problem. The following bound states explicitly the distributional assumption needed for a margin interpretation and assigns ties to the decision not to mix.

Proposition S2.1 (margin law for a threshold rule)**.** *Let $D = \varrho - e_1/e_2$ be the gap in Proposition 6.1(i), and define the binary decision $d(x) = \mathbf{1}\{x < 0\}$. Thus $d(D) = 1$ means that mixing strictly helps, and $d(0) = 0$. Suppose, conditionally on D, an estimate $\widehat{D}$ satisfies, for some fixed $\sigma > 0$,*

$$\mathbb{E}\Big[\exp\{t(\widehat{D} - D)\} \mid D\Big] \le \exp(t^2\sigma^2/2), \qquad t \in \mathbb{R}.$$

Then:

(i) for $D \neq 0$, writing $\Delta = |D|$, $\mathbb{P}(d(\widehat{D}) \neq d(D) \mid D) \le \exp(-\Delta^2/(2\sigma^2))$;

(ii) for every $\tau \ge 0$, including when $D = 0$,

$$\mathbb{P}\big(d(\widehat{D}) \neq d(D),\ |\widehat{D}| \ge \tau\big) \le \exp(-\tau^2/(2\sigma^2)).$$

Proof Condition on D. If $D < 0$, a wrong decision requires $\widehat{D} \ge 0$, hence $\widehat{D} - D \ge |D|$. If $D > 0$, a wrong decision requires $\widehat{D} < 0$, hence $D - \widehat{D} > |D|$. The one-sided Chernoff bound obtained by minimizing $\exp(t^2\sigma^2/2 - t\Delta)$ over $t > 0$ gives (i). For (ii), when $D < 0$ the event in question implies $\widehat{D} \ge \tau$, so $\widehat{D} - D \ge \tau$. When $D \ge 0$, including a tie, it implies $\widehat{D} < 0$ and $\widehat{D} \le -\tau$, so $D - \widehat{D} \ge \tau$. The same one-sided bound gives $\exp(-\tau^2/(2\sigma^2))$ in either case, and averaging over D proves (ii). At $\tau = 0$ the bound is the trivial upper bound one. ■

Part (ii) bounds the joint event of a wrong decision and a large observed margin. It does not bound the error rate conditional on selecting such a margin; that would require division by $\mathbb{P}(|\widehat{D}| \geq \tau)$. The hypothesis also implies conditional centering of the estimation error and a sub-Gaussian tail bound, neither of which follows from a small pooled mean or standard deviation. The differences across the 840 dependent member pairs describe the observed validation-to-test transfer scale, not a certified value of σ. In particular the OCO-2 validation and test blocks have the distribution shift described in Section 5. The margin-bin accuracies and the illustrative exponential calculation of Section 5.2 are therefore empirical diagnostics and a conditional reference calculation, respectively.

A separate observation concerns separable squared output metrics. It motivates coordinatewise combinations without implying simultaneous optimality for the two reported means of relative norms.

Proposition S2.2 (per-coordinate combination under diagonal metrics)**.** *For weights $v = (v_{mj})$ that may depend on the output coordinate, let $f_v(u)_j = \sum_m v_{mj} f_m(u)_j$. For any diagonal metric with positive weights w,*

$$\mathbb{E}\big\|\mathrm{diag}(w)\big(f_v(u) - G(u)\big)\big\|_2^2 = \sum_j w_j^2\, \mathbb{E}\big(f_v(u)_j - G(u)_j\big)^2,$$

so the minimizing weights solve q separate problems, one per output coordinate, none of which involves w. The per-coordinate optimum is the same for every diagonal metric, and it weakly dominates every combination whose weights are shared across coordinates, for all such metrics simultaneously.

Proof. See Supplement S8.14.

Here each coordinate's feasible weights may be chosen independently, and the optimum is a population optimum. With fixed positive diagonal weights, the squared risk separates in this way. An empirical minimizer has the same algebraic separation for the empirical objective, but need not be a population minimizer. A common per-sample weight also preserves separation.

Corollary S2.3 (weighted metrics and the reported error)**.** *Let $a : \mathcal{U} \to [0, \infty)$ be measurable with $\mathbb{E}[a(u)\, \|f_v(u) - G(u)\|_2^2] < \infty$. Then, for every diagonal w,*

$$\mathbb{E}\Big[a(u)\, \big\|\mathrm{diag}(w)\big(f_v(u) - G(u)\big)\big\|_2^2\Big] = \sum_j w_j^2\, \mathbb{E}\Big[a(u)\, \big(f_v(u)_j - G(u)_j\big)^2\Big],$$

so the minimization still splits into q per-coordinate problems, none involving w. Taking $a(u) = \|G(u)\|_2^{-2}$ and $w = \mathbf{1}$ makes the left side the mean squared relative error: the population combination minimizing each weighted coordinate risk is simultaneously optimal, within its class, for the squared relative error and for every diagonal reweighting of it. The reported metric, $\mathbb{E}\,\|\rho_{f_v}\|_2$ with ρ the normalized residual of Section 6.2, is controlled through $\mathbb{E}\|\rho\|_2 \leq (\mathbb{E}\|\rho\|_2^2)^{1/2}$; the gap between the two sides is a predictor-specific dispersion factor. For the fitted mechanics mixtures it is about 0.94 *on validation; it must be measured separately for other predictors and samples.*

Proof. See Supplement S8.15.

The experiments run the combination both ways, unweighted and with the per-sample weights of the corollary, and report both.

Proposition S2.2 and its corollary describe the population optimum; the pipeline computes an empirical selection, coordinate by coordinate, on the validation split. For candidates fixed independently of an i.i.d. validation sample, the following inequality bounds its finite-sample selection cost.

Proposition S2.4 (empirical per-coordinate selection)**.** *Let $f_1, \dots, f_M$ be fixed maps, and let $a : \mathcal{U} \to [0, A]$ be a bounded per-sample weight. Suppose $a(u)\,(f_m(u)_j - G(u)_j)^2 \le L$ for μ-a.e. u, every member m and coordinate j. Take a fixed independently of validation. On an i.i.d. validation sample from μ of size m_v independent of the members, let $\widehat{m}(j)$ minimize the empirical weighted risk of coordinate j over the M members, and let the combination take coordinate j from member $\widehat{m}(j)$. Then, with probability at least $1-\delta$, simultaneously for every coordinate,*

$$R_j\big(\widehat{m}(j)\big) \le \min_{m \le M} R_j(m) + 2L\sqrt{\frac{\log(2Mq/\delta)}{2m_v}},$$
$$R_j(m) = \mathbb{E}\big[a(u)\big(f_m(u)_j - G(u)_j\big)^2\big].$$

and consequently, for every diagonal metric w at once,

$$\sum_j w_j^2 R_j(\widehat{m}(j)) \le \sum_j w_j^2 \min_m R_j(m)$$
$$+ 2L\Big(\sum_j w_j^2\Big)\sqrt{\frac{\log(2Mq/\delta)}{2m_v}}.$$

Proof. See Supplement S8.16.

The OCO-2 selector uses this coordinatewise empirical operation with $M = 6$, $q = 40$ and $m_v = 2000$. Its candidate checkpoints and kernel heads were themselves selected on the same validation block, and that block has a different distribution from the test block. The proposition therefore does not provide a risk guarantee for that deployed selector. Its observed gains on both reported metrics, and the comparison of unweighted and inverse-norm-weighted selectors, are empirical findings.

S2.1 A finite-design bound for truncating an input map

An approximate rank alone does not imply a learning rate. The next result instead separates a deterministic approximation error from the kernel error on a specified lower-dimensional design.

Proposition S2.5 (finite-design rank truncation)**.** *Let $\mathcal{U} \subset \{u \in \mathbb{R}^d : \|u\|_2 \le R\}$ and let $G(u) = g(Au)$, where $A \in \mathbb{R}^{d\times d}$ has singular values $\sigma_1 \ge \cdots \ge \sigma_d \ge 0$ and $g : \mathbb{R}^d \to \mathbb{R}^q$ is L-Lipschitz in Euclidean norm. For $1 \le r \le d$, write $A_r = U_r \Sigma_r V_r^\top$ for a truncated singular-value decomposition, put $\sigma_{d+1} = 0$, and define*

$$\varphi_r(u) = \Sigma_r V_r^\top u, \qquad h(z) = g(U_r z), \qquad G_r = h \circ \varphi_r, \qquad \varepsilon = LR\sigma_{r+1}.$$

Suppose the components of h belong to the RKHS $\mathcal{H}_k$ of a positive definite kernel on $\mathbb{R}^r$, with $\|h\|_K^2 = \sum_{j=1}^q \|h_j\|_{\mathcal{H}_k}^2 < \infty$. For any design $X = (u_1, \dots, u_n)$ in $\mathcal{U}$, let $Z =$

$(\varphi_r(u_1),\ldots,\varphi_r(u_n))$, $K = k(Z,Z)$ and

$$c_\lambda(u) = (K + n\lambda I)^{-1} k(Z, \varphi_r(u)), \qquad \widehat{G}_\lambda(u) = c_\lambda(u)^\top G(X),$$

where $\lambda \ge 0$, and at $\lambda = 0$ the inverse is interpreted as a pseudoinverse when necessary. Let $\widetilde{P}_\lambda(u)$ be the factor in Theorem 6.9, evaluated in this feature space; equivalently,

$$\widetilde{P}_\lambda(u) = \left\| k(\varphi_r(u),\cdot) - \sum_{i=1}^{n} c_{\lambda,i}(u) k(\varphi_r(u_i),\cdot) \right\|_{\mathcal{H}_k}.$$

Then, for every $u \in \mathcal{U}$,

$$\|G(u) - \widehat{G}_\lambda(u)\|_2 \le \|h\|_K \, \widetilde{P}_\lambda(u) + \varepsilon\big(1 + \|c_\lambda(u)\|_1\big).$$

Proof. See Supplement S8.18.

The proof is in Supplement S8.18. The first term is the native-space bound on the projected design. The second includes both the pointwise approximation error from discarding directions and its propagation through the regression weights. For $\lambda > 0$ and bounded kernel diagonal, Lemma S1.7 also bounds the latter weights uniformly. No sampling rate or comparison with the unprojected kernel follows without additional control of the power function and the target norm. The empirical OCO-2 scaling sweep reports those observed error curves without treating the measured sensitivity rank as such a control.

S3 Coverage of calibrated prediction balls

S3.1 Coverage under split-conformal assumptions

The native-space bound of Theorem 6.9 contains an unknown norm, and P_λ ranks the measured pointwise errors weakly. A separate calibration construction rescales this positive score by an empirical quantile. Write $\widehat{G}$ for the fitted predictor, $e(u) = G(u) - \widehat{G}(u)$, and let

$$s_i = \frac{\|e(\tilde{u}_i)\|_2}{P_\lambda(\tilde{u}_i)}, \qquad k = \lceil (1-\alpha)(m+1) \rceil, \qquad Q = \begin{cases} s_{(k)}, & k \le m, \\ +\infty, & k = m+1, \end{cases} \tag{S3.1}$$

where $s_{(k)}$ is the k-th order statistic of the m calibration scores and $0 < \alpha < 1$. When the required index exceeds the calibration sample size the prediction set is the whole output space.

Proposition S3.1 (finite-sample coverage)**.** *Fix the predictor $\widehat{G}$, design X and positive scale function P_λ. Suppose the calibration and test inputs $\tilde{u}_1,\ldots,\tilde{u}_m, u^\star$ are exchangeable and independent of these fitted objects. Define Q by* (S3.1) *and*

$$C(u) = \{v : \|v - \widehat{G}(u)\|_2 \le Q P_\lambda(u)\}.$$

Then $\mathbb{P}(G(u^\star) \in C(u^\star)) \ge 1-\alpha$. If the scores are almost surely distinct, their coverage is exactly $k/(m+1)$, and hence is at most $1-\alpha+1/(m+1)$. When $k = m+1$, the coverage is one, regardless of ties.

Proof. See Supplement S8.21.

Remark S3.2 (conditional reference law for realized coverage)**.** *Suppose, more strongly, that conditional on the fixed fitted predictor and scale function, calibration and future evaluation scores are i.i.d. from a continuous distribution F. If $1 \le k \le m$, the coverage probability conditional on the calibration sample is $F(s_{(k)})$. The probability integral transform makes $F(s_i)$ i.i.d. uniform, so*

$$F(s_{(k)}) \sim \mathrm{Beta}(k, m+1-k).$$

This is the standard order-statistic law (Vovk, 2012). Conditional on the calibration sample, the count of covered cases in an independent evaluation set of size N is binomial with success probability $F(s_{(k)})$; integrating gives a Beta–binomial count. For $k = m+1$ the conditional coverage is instead identically one and the count is N; there is no Beta distribution with a zero second parameter. Exchangeability alone is sufficient for Proposition S3.1, but does not imply this stronger Beta–binomial law.

The proposition is the usual split-conformal rank argument (Vovk et al., 2005; Lei et al., 2018); its proof is in Supplement S8.21. The score is a residual norm divided by a positive scale, and the resulting set is a Euclidean ball in output space with radius $QP_\lambda(u)$. The stored statistic named `mean_width` in the records is the mean of this radius, not its diameter. The guarantee is marginal over a new input, not conditional coverage at each input and not a collection of coordinatewise intervals. Constant-radius and other positive-scale scores satisfy the same proposition under the same independence assumptions, whether or not the scale ranks error well.

The released campaign applies this calculation to a 1000/19000 partition of the public test block after the reported models were fitted. Its recorded six-member coverage is $89.9 \pm 0.8\%$ at the nominal 90% level, with P_λ rebuilt from each seed's correction kernel (`campaign/uq_conformal_plam.py`). Under the i.i.d.-continuous reference model, $m = 1000$, $\alpha = 0.1$ give $\mathrm{Beta}(901, 100)$ conditional coverage, and the central 95% reference interval for the fraction on $N = 19000$ cases is $[88.0\%, 91.8\%]$; at the 95% nominal level it is $[93.5\%, 96.3\%]$. The reported ten seeds fall inside these intervals at both levels. These are conditional reference comparisons. The seeds share test cases and are not ten independent draws from this law.

The proposition's guarantee refers to a predictor and scale fixed before an independent, exchangeable calibration-and-evaluation sample is observed. The measured coverage, the constant-radius comparison and the radius ratios are empirical results. Under the stated fresh-sample assumptions, the conformal guarantee does not require native-space membership, a known residual norm, or a positive association between P_λ and error.

S4 The rest of the suite, and the mirror-symmetry check

S4.1 The remaining problems of the suite: an uncontrolled survey

For orientation we also ran one fixed recipe, unchanged, on the other problems of the operator-learning suite that Batlle et al. (2024) assembled from de Hoop et al. (2022) and Lu et al. (2022): a Matérn kernel at the median length scale with the fit capped at 6000 points, a residual MLP mean, outputs reduced by PCA, and the stage selected on validation. Table S4.1 lists the results next to the reference values collected in that study. The comparison is not controlled, and we draw nothing from it beyond orientation: the

Table S4.1: The remaining problems of the suite: an uncontrolled survey. Published reference mean relative L^2 test error on each problem (per-problem methods tuned by their authors), against one recipe run unchanged, on training splits and grids that differ from the baselines' in the cases noted in the text: single runs, except the two rows marked three seeds (mean $\pm$ standard deviation, exact solve on 19,000 pairs, output rank 400).

Problem	Map	Published reference	This work
Burgers	initial condition $\to$ solution	1.93% (FNO)	2.38%
Darcy flow	permeability $\to$ pressure	2.32% (POD-DeepONet)	2.01%
Advection I	smooth pulse $\to$ solution	$\approx$ 0% (kernel)	—
Advection II	discontinuous pulse $\to$ solution	11.28% (linear kernel)	11.29%
Helmholtz	wavespeed $\to$ field	1.00% (kernel)	1.17% $\pm$ 0.02 (three seeds)
Navier–Stokes	vorticity $\to$ vorticity	0.12% (kernel)	0.28% $\pm$ 0.00 (three seeds)

published numbers are per-problem methods tuned by their authors; our rows are single runs without seed replication, except Helmholtz and Navier–Stokes, which were rerun at three seeds with the exact solve on 19,000 training pairs once it was found that the recipe's output rank of 40, adequate for the stress fields, left a reconstruction floor of 1.7% and 8.2% on these two problems (the rerun uses rank 400, whose floor is below 0.01%, and every rerun head sits well above it); Burgers and Darcy use slightly smaller training splits than their baselines (800 and 824 pairs) and Darcy a different grid, from the copies we could obtain; and we report no number for Advection I. What the survey shows is the sorting one would expect: where the map is smooth and data are plentiful (Helmholtz, Navier–Stokes, Darcy) the kernel stage wins the recipe's internal selection and lands within a factor of 1.2 (Helmholtz) to 2.3 (Navier–Stokes) of the tuned per-problem kernels, whose Cholesky preconditioning of the input the recipe does not use, and on Burgers the corrected mean sits near the Fourier neural operator. On Helmholtz the per-coordinate selection improves on the kernel alone (1.17% against 1.22%) by handing a few dozen of the 400 output coordinates to the corrected mean; on Navier–Stokes it selects the kernel almost everywhere. The two problems studied in depth, with replication, are the subject of the rest of the paper.

S4.2 A data-driven check of the mirror symmetry

The continuous problem is invariant under $x_1 \mapsto 1 - x_1$: the domain and boundary partition are symmetric, and the input law has a symmetric covariance. On the grid, reflecting the load (S) should reflect the stress field along the first grid coordinate (T), i.e. $G(Su) = TG(u)$. Rather than assume this, we test it on the data. For each of the first 200 samples we searched all 40000 loads for the nearest neighbor of the reflected load Su_i and compared the corresponding outputs. For the five best-matching pairs, the input mismatch $\|Su_i - u_j\|/\|u_j\|$ ranges over 0.27–0.32, while the output mismatch $\|Tv_i - v_j\|/\|v_j\|$ ranges over 0.085–0.14; reflecting along the second coordinate instead gives output mismatches of order one. Outputs of near-mirror inputs are far closer than the inputs themselves, which is what equivariance plus a Lipschitz solution map predicts, and the effect singles out the correct output reflection axis. A complementary check on the input law: the empirical mean and covariance of the training loads are invariant under reflection to within 1.1% and 1.5% respectively. These moment checks are consistent with, but do not establish, the exact input-law invariance assumed by Proposition S1.1. Consistently with this evidence, reflection averaging at test time improves twenty-nine of the thirty reflected members of the seed campaign (Section 4). This direction is consistent with the symmetry motivation. For the true full-map average, Proposition S1.1 controls expected loss under its exact assumptions; a finite test sample can go the other way. The thirty records include ten refiners whose supplied-field averaging requires a further channel-equivariance condition, as explained in Supplement S1.1; their improvements are empirical support for that implementation, not instances of the proposition's guarantee.

S5 Effective dimension from the Gram spectrum

S5.1 Effective dimension from the Gram spectrum

The remaining design choice is the nugget λ, selected by cross-validation in the pipeline. The following exact identity connects that choice to the spectrum of the Gram matrix and, through it, to random matrix descriptions of the design.

Lemma S5.1 (effective dimension and the empirical Stieltjes transform)**.** *Let K be a symmetric positive semidefinite $n \times n$ matrix with eigenvalues $\lambda_1, \dots, \lambda_n \geq 0$, let $\widehat{m}(z) = \frac{1}{n}\sum_{i=1}^{n}(\lambda_i/n - z)^{-1}$ be the Stieltjes transform of the empirical spectral distribution of K/n. Then for every $\lambda > 0$ the effective dimension of ridge regression with nugget λ satisfies*

$$d_{\mathrm{eff}}(\lambda) \;:=\; \operatorname{tr}\big(K(K + n\lambda I)^{-1}\big) \;=\; n\big(1 - \lambda\,\widehat{m}(-\lambda)\big).$$

In particular, if the empirical spectral distribution of K/n converges weakly to a limit law F as $n \to \infty$, then $d_{\mathrm{eff}}(\lambda)/n \to 1 - \lambda\, m_F(-\lambda)$ with m_F the Stieltjes transform of F.

Proof. See Supplement S8.23.

The identity is unconditional; the random-matrix content enters through the choice of F. For the benchmark's simplest reference model, isotropic linear features, the limit can be evaluated in closed form.

Corollary S5.2 (effective dimension under a Marčenko–Pastur limit)**.** *Let $K = XX^\top$, where $X \in \mathbb{R}^{n\times p}$ consists of the leading entries of an infinite array of i.i.d. real random variables with mean zero and finite variance $\sigma^2 > 0$. Let $p/n \to \gamma \in (0, 1]$. For each fixed $\lambda > 0$,*

$$\frac{d_{\mathrm{eff}}(\lambda)}{n} \;\longrightarrow\; \gamma\big(1 - \lambda\, m(-\lambda)\big),$$

where

$$m(-\lambda) = \frac{\sqrt{\big(\lambda + \sigma^2(1-\gamma)\big)^2 + 4\gamma\sigma^2\lambda} - \big(\lambda + \sigma^2(1-\gamma)\big)}{2\gamma\sigma^2\lambda},$$

almost surely, where m is the Stieltjes transform of the Marčenko–Pastur law with ratio γ and scale σ^2.

Proof. See Supplement S8.22.

In one Gaussian realization with $n = 3000$, $p = 900$, and $\sigma^2 = 1.7$, the largest absolute difference between the limit formula and the empirical $d_{\mathrm{eff}}(\lambda)/n$ was 4.4×10^{-5} over the five values $\lambda \in \{10^{-3}, 10^{-2}, 10^{-1}, 1, 10\}$.

The closed form is a reference for the stated i.i.d. linear-feature model. The learned neural features and nonlinear kernel matrices in the experiments lie outside those assumptions; the finite-matrix trace identity holds regardless. Figure S6.3 applies it to a separate 6000-row subsample of the structural-mechanics training design. At the diagnostic's fixed nugget 10^{-5}, its effective dimension is 103.12. This is the trace of that subsample smoother, not a measurement of the full 19000-row deployed fit. The trace neither certifies generalization nor reduces the cubic cost of a dense factorization.

The effective dimension also determines the sum of the training-point posterior variances. Writing $A = K + n\lambda I$, their sum is

$$\sum_{i=1}^{n} P_\lambda(u_i)^2 = \operatorname{tr}(K) - \operatorname{tr}(KA^{-1}K) = \operatorname{tr}\big(n\lambda\, KA^{-1}\big) = n\lambda\, d_{\mathrm{eff}}(\lambda),$$

using $KA^{-1}K = K - n\lambda\, KA^{-1}$. This identity concerns the training design. The trace alone does not determine the power function at other inputs.

S6 Additional structural-mechanics results

S6.1 Ablation

Table S6.1 builds the pipeline up one component at a time. The kernel ridge on loads and the reflection-averaged neural mean start at 5.197% and 4.836%. Stacking is trivial with a single high-data mean and leaves it unchanged; the value of the combination appears once there are several means to combine and again in the correction stage. The feature-space and second input-space corrections did not improve validation error beyond the first correction with a single mean and so were not selected; we report the stages that the validation split actually chose.

Reflection test-time averaging lowers the metric-trained MLP's test error by 0.16 points. This agrees with the direction of Proposition S1.1 under its symmetry assumptions; the empirical symmetry checks do not prove those assumptions. The effect is smaller on the normalized-MSE network (0.02 points). In the ten-seed refiner records, the implemented supplied-field average lowers test error by 0.0422 ± 0.0019 percentage points (mean and sample standard deviation), with an improvement at every seed. Random reflection augmentation during training does not make the averaging effect vanish. The additional equivariance condition of Proposition S1.1 is not assumed for this refiner; these changes are empirical measurements. The kernel-flow regularizer did not improve the low-data MLP's validation error. At high data the ten-seed records give 4.99% at weight 0.3 against 4.99% without it, and 5.05% at weight 1.0 in the smaller campaign.

Replacing the single tuned Matérn correction by a sum at several bandwidths leaves validation error unchanged to two decimals. This is a result for the searched kernels and tuning budget, not an irreducibility claim about the residual. In the five-member campaign, nine seeds select the largest nugget on the grid, $\lambda = 10^{-3}$, and one selects 10^{-5}. The length-scale multiplier is 2 at five seeds, 1 at four and 4 at one. The four-member ablation selects multiplier 2 at eight seeds. These choices describe the regularization preferences of the fitted corrections.

Table S6.1: Building the high-data pipeline. Mean relative L^2 test error under the 20000-sample protocol; "+" rows are cumulative. All rows are mean ± standard deviation over ten seeds; the last two are the second campaign of Section 4.1.

Stage	Test error (%)
Kernel ridge on loads	5.197 ± 0.003
Reflection-averaged neural mean	4.836 ± 0.018
+ refiner and MSE variant, affine stack	4.628 ± 0.009
+ residual kernel correction	4.611 ± 0.005
+ FNO member, affine stack	4.595 ± 0.008
+ residual kernel correction	4.572 ± 0.010
+ complete FNO schedule and UNet member, affine stack	4.567 ± 0.004
+ residual kernel correction	4.546 ± 0.003

The kernel prediction supplies the refiner's conditioning field. Its residual is less correlated with the neural members' residuals on the measured splits, which motivates this input. Section 4.5 reports the ensemble comparisons and their second-moment diagnostics.

S6.2 Where the remaining error is

The error measurements describe both the size and the location of the remaining residual within the trained pool. Ranked by relative error on the validation split, the twenty hardest cases have error 10.24% for the reference member against a median of 4.44%, a factor of 2.31. On those same cases the five-member ensemble reaches 10.18% and the mean over individual members is 10.28%. The ensemble improves this tail only slightly. Because these cases were selected by the reference member's errors, the comparison describes that selected subset rather than an independent tail-risk experiment.

The root-mean-square pointwise error over validation samples is 8.99 on the perimeter of the 41×41 grid and 5.64 in the interior, a ratio of 1.59; both vary across seeds by less than one percent of their values. Averaging lowers them to 8.87 and 5.59. The concentration near the loaded edge and supports remains visible in Figure 1. These measurements identify shared spatial structure in the predictors' errors. They do not distinguish approximation bias from numerical artifacts in the target fields.

S6.3 Spectra and effective dimension

Figure S6.3 reports a separate diagnostic on 6000 rows sampled without replacement from the 19000 training rows. The load coordinates are standardized using the full training design. The Matérn-5/2 length scale is fixed at four times the median distance on a 2000-row training subsample, and the marked nugget is 10^{-5}. These match the parameters recorded for the earlier hybrid diagnostic; they are not a new selection or the parameters of every deployed correction. The split and sampling generators both use seed 0.

The leading eight eigenvalues carry 99% of the trace, while $d_{\mathrm{eff}}(10^{-5}) = 103.11885$. Spectral mass and effective dimension summarize different aspects of the matrix: the latter counts each mode with weight $\mu_i/(\mu_i + 6000\lambda)$. An independent Cholesky calculation of

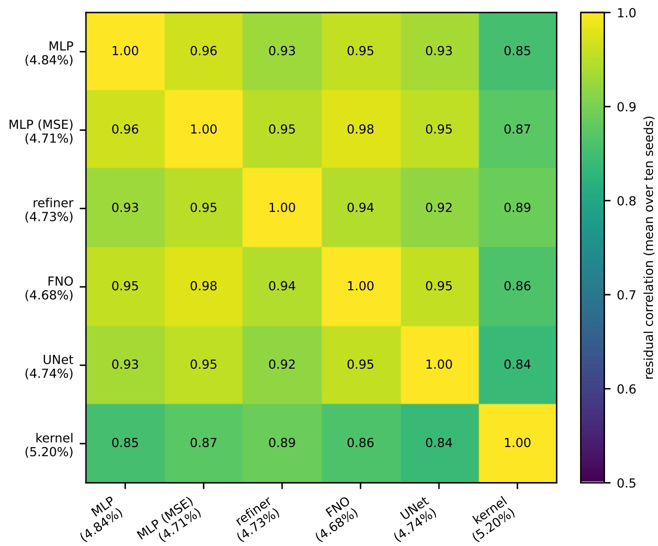

Figure S6.1: Correlation of per-sample normalized residuals on the validation split, six members, averaged over the ten seeds of the second campaign. Across seeds and pairs the entries range between 0.82 and 0.98, with a mean of 0.918 ± 0.005; the kernel predictor still correlates 0.86–0.88 with the most accurate network at every seed.

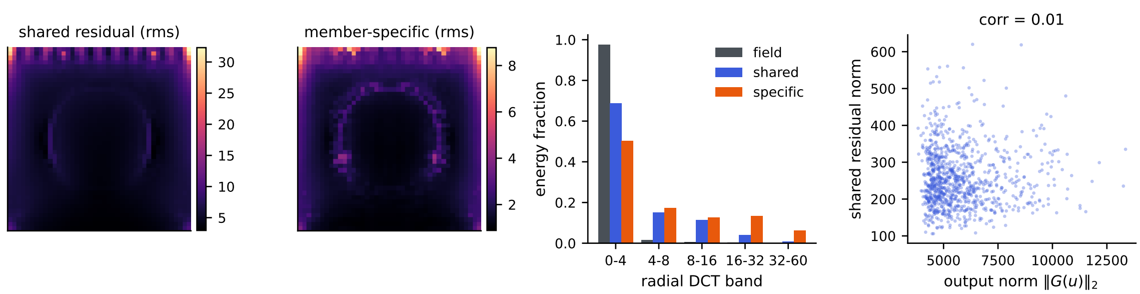

Figure S6.2: Anatomy of the shared residual c (across-member mean of the residual fields), at one seed. Left to right: rms of c over the grid; rms of the member-specific remainders; radial DCT band energies of the stress fields, of c, and of the specific parts; and the norm of c against the output norm, whose correlation is 0.01.

$6000 - 6000\lambda \operatorname{tr}\left((K_S + 6000\lambda I)^{-1}\right)$ agrees with the eigenvalue sum to 1.7×10^{-11}. The directory `campaign/collected/spectrum_20260906` retains the selected rows, normalization, eigenvalues and generation record. This regenerated diagnostic does not measure the full deployed smoother. Its spectral concentration also does not reduce the cubic work of the dense factorization used here.

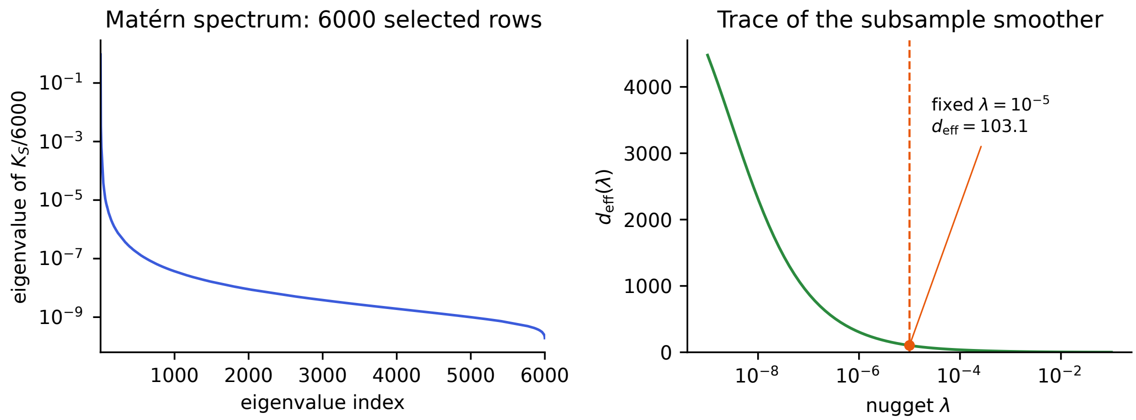

Figure S6.3: A regenerated 6000-row training-subsample diagnostic with the fixed kernel parameters specified in the text. Left: eigenvalues of $K_S/6000$ on a log scale. Right: the effective dimension from Lemma S5.1; the marker at the fixed nugget 10^{-5} gives 103.12. Both panels describe this subsample matrix. The eigenvalues and an independent direct trace check are included in the numerical record.

S6.4 Cost and reproducibility

The pipeline was developed on a single workstation GPU and replicated on four cloud CPU instances. The kernel stages use dense, exact linear solves. At the campaign's shapes, an eight-thread timing diagnostic measured 27 seconds to build a 19000×19000 Gram matrix in blocks, 22 seconds for its Cholesky factorization, and 11 seconds to solve for 1681 output coordinates. The roughly one-minute total excludes hyperparameter search, network training and prediction. Values in this timing experiment are synthetic; the dimensions and operations match the campaign. The factorization performs $n^3/3 \approx 2.3 \times 10^{12}$ floating-point operations. Low effective dimension describes the regularized spectrum but does not reduce this dense factorization's arithmetic cost. No GPU, inducing points, preconditioner or stochastic linear solver is used in the correction.

Building the Gram matrix, not factoring it, is the stage that needs care at this size: forming it through a full squared-distance matrix and its elementwise transforms allocates several 2.9 GB temporaries at once, which exhausted a 31 GB instance during the campaign. The block-filled timing diagnostic has a measured peak of 6.0 GiB, including the factorization and solves, compared with 19.2 GiB after the full-matrix construction. This is a demonstrated implementation option: the main correction driver still forms full-size temporaries and needs the corresponding memory. The neural means have a few million parameters and include both metric-loss and normalized-MSE training. Splits are specified by permutation seeds and inputs are the 41-dimensional loads described in Section 2. The fitting stages use validation-based choices. The release contains splits, code and per-seed summaries for checking reported rows. Prediction arrays and checkpoints are not distributed, so regenerating predictions requires retraining.

S6.5 Uncertainty quantification

Theorem 6.9 gives a conditional bound for a residual function in the kernel native space with consistent training labels. The deployed correction uses mixed training predictions: the kernel channel is constructed with foldwise fits whose target centering uses the pooled training labels, while neural predictions on these rows are in-sample. These labels need not equal $G(X) - m_{\text{full}}(X)$ for the deployed mean. The theorem applies to the deployed error under its stated consistency and native-space assumptions and, where applicable, the mismatch term.

Table S6.2 reports a computable diagnostic: the squared RKHS norms of ridge functions fitted to the two label arrays. With the selected correction kernel at seed 0, the residual-label fit has a norm about 4900× smaller than the raw-stress fit and about 8.8× smaller per unit label energy. These ratios depend on the nugget and the scale. The mixed residual labels prevent identifying this fitted norm with a lower bound for the native-space norm of the deployed residual $G - m_{\text{full}}$. No such lower bound or smoothness conclusion is claimed.

The design factor P_λ correlates weakly with the deployed surrogate's absolute error (Pearson 0.055) and negatively with relative error. Its correlation with the output norm is 0.65 (Figure S6.4), so normalization by output amplitude changes how this signal should be read. These associations describe the measured data. Ensemble disagreement has correlation 0.074 ± 0.006 with absolute error across ten seeds and -0.25 ± 0.04 with relative error. The corresponding four-member absolute error correlation is 0.083. Thus neither signal ranks the remaining error well. A spread statistic is insensitive to a common additive error shared by all members, which is consistent with the residual alignment observed here.

Three bands are calibrated on 1000 cases selected from the test block and scored on the other 19000: the scores are $\|e\|_2/P_\lambda$, $\|e\|_2$ (constant radius), and $\|e\|_2$ divided by ensemble disagreement. Each seed's P_λ uses its own selected correction kernel and training design. These cases are disjoint for this calibration exercise. The measured coverages below are split diagnostics on the public test block. The marginal split-conformal theorem applies when the predictor and score are fixed independently of exchangeable calibration and future observations; the exact Beta and Beta–binomial descriptions additionally require conditionally i.i.d. continuous scores (or an appropriate randomized rank construction). They are not consequences of a random split of an arbitrary fixed benchmark pool.

For the six-member pipeline the P_λ band has empirical coverage $89.9 \pm 0.8\%$ at the nominal 90% level and $95.0 \pm 0.6\%$ at 95%. Under the i.i.d.-continuous reference model, $m = 1000$ gives latent coverage $\text{Beta}(901, 100)$ at the 90% level; with 19000 independent evaluation scores its Beta–binomial reference interval is $[88.0\%, 91.8\%]$. The analogous 95% interval is $[93.5\%, 96.3\%]$. All ten six-member coverages fall inside these intervals at each level.

The constant-radius band covers $90.4 \pm 0.8\%$ and $95.4 \pm 0.5\%$, inside the reference intervals at ten of ten seeds at both levels; the P_λ band is on average 1.11 times as wide. The disagreement-scaled band covers $90.1 \pm 0.4\%$ and $95.0 \pm 0.8\%$, with one seed on the upper edge of the 95% reference interval. The five-member pipeline gives $90.1 \pm 0.9\%$ and $95.1 \pm 0.8\%$ for the P_λ band (ten and nine seeds inside), and $90.6 \pm 0.4\%$ and $95.4 \pm 0.6\%$ for the constant-radius band (ten inside at both levels). The records are `collected/dgx/uq_plam_seeded.json` and the `uq` field of each seed's pipeline record, pro-

Table S6.2: Fitted-norm diagnostic of the residual correction at seed 0 of the complete-schedule campaign: the same validated correction kernel (scale multiplier 2, nugget 10^{-3}, median length scale) ridge-fitted to the raw stress fields and to the six-member stack residuals on the training split. $\|\widehat{r}\|_K^2 = \mathrm{tr}(\alpha^\top K \alpha)$ is the squared RKHS norm of the fitted function; the last column divides it by the energy of the fitted label array. The residual labels mix foldwise kernel predictions and in-sample neural predictions, so these numbers are not identified as lower bounds for the native-space norm of the deployed residual. The ratios depend on the nugget and the scale the correction selected, and most of the raw ratio is amplitude: per unit energy the residual's fitted norm is 8.8 times smaller at this seed, not 4900; at seeds 1 and 2, whose corrections chose scale multipliers 1 and 2, the raw ratios are 1800 and 4300 and the normalized ones 3.2 and 7.6. Records `collected/dgx/sm_norm_check_hpix_s{0,1,2}.json`.

Kernel regresses	$\|\widehat{r}\|_K^2$	energy $\|T\|_F^2$	$\|\widehat{r}\|_K^2/\|T\|_F^2$
Raw stress fields ($m=0$)	7.12×10^8	6.77×10^{11}	1.05×10^{-3}
Ensemble residuals	1.45×10^5	1.21×10^9	1.20×10^{-4}
Ratio	4900×	560×	8.8×

duced by `campaign/uq_conformal_plam.py` and `campaign/conformal_seeded.py`. Here the constant-radius band is narrower, and the kernel design factor provides no measured efficiency gain.

S6.6 Which uncertainty is the binding one

The seed standard deviations describe variation from initialization and recorded selection on a common fixed test block. They do not include the sampling uncertainty of a future evaluation set. If cases were independent and representative of a target population, the reported per-case standard deviation 0.0169 would give a mean-error standard error of 0.012 percentage points over 20000 cases. We retain this as a sampling-scale calculation under those assumptions.

The published baselines do not provide per-case predictions or comparable training replicates. Their differences from our scores therefore do not support a paired significance test or an equivalence conclusion. Combining two copies of our sampling term in quadrature gives approximately 0.017 points, but this imputes an unknown baseline variance and covariance structure. It is only a sensitivity calculation, not an estimated standard error for the difference. The 0.098-point difference from PCA-Net and 0.022-point difference from PARA-Net for the five-member pipeline are benchmark-score differences; neither is declared statistically significant, equivalent or conservative on that basis.

Within the two campaigns, per-case paired differences isolate changes between fitted predictors on the same cases. The end-to-end gain is 0.140 points and removal of the FNO costs 0.039 ± 0.011 points across ten seeds, with the latter direction holding at all ten. These describe repeatability across seeds and paired test-block effects.

The validation second-moment simplex optimum distributes weight among the five members: refiner 0.31 ± 0.04, FNO 0.34 ± 0.06, normalized-MSE network 0.24 ± 0.08,

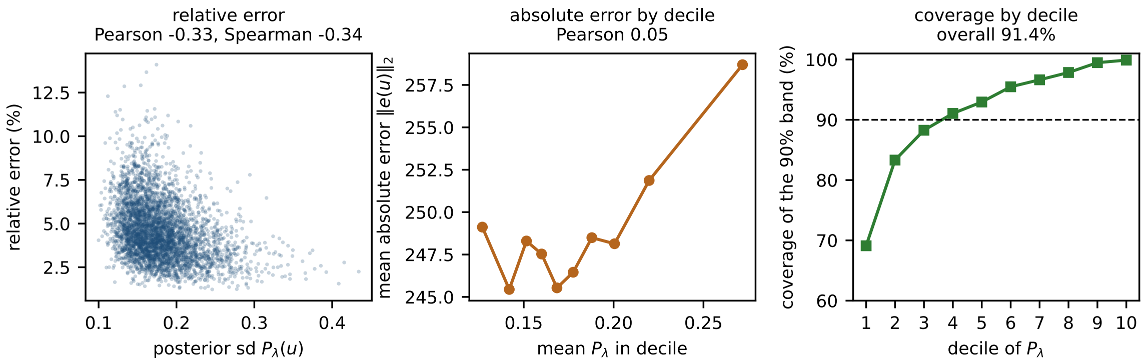

Figure S6.4: One seed (seed 0), the deployed six-member correction, on the evaluation subset of the test block (19000 cases) with the band calibrated on the other 1000 as in Section 4.7. Left: the Gaussian process posterior standard deviation P_λ against relative error; the association is negative (Pearson -0.33) because large-amplitude outputs have small relative error. Middle: decile calibration of P_λ against absolute error $\|e(u)\|_2$; the association is weak (Pearson 0.05). Right: coverage of the P_λ-scaled 90% band within each decile of P_λ: 91.4% overall, but from 69% in the lowest decile to 99.9% in the highest, so coverage varies strongly with the design factor despite an overall value close to the nominal level. Across the ten seeds the same band records $89.9 \pm 0.8\%$ at the nominal 90% level, inside the i.i.d.-continuous Beta–binomial reference interval at ten seeds of ten; the constant-radius band is the narrower of the two (Section 4.7).

kernel 0.06 ± 0.03, and metric-trained MLP 0.06 ± 0.04. The kernel has nonzero weight at all ten seeds and the metric-trained MLP at nine. For the six-member campaign the corresponding shares are UNet 0.32 ± 0.06, FNO 0.30 ± 0.07, refiner 0.27 ± 0.04, kernel 0.06 ± 0.02, normalized-MSE network 0.03 ± 0.04, and metric-trained MLP 0.02 ± 0.02. The FNO's large share despite its single-member ranking illustrates the role of off-diagonal residual second moments.

Proposition 6.1(i) uses the uncentered alignment $S_{12}/\sqrt{S_{11}S_{22}}$. The descriptive centered correlations in Figure S6.1 and the early admission summaries are not automatically interchangeable with that alignment. We therefore use the matrix objective for exact convex-mixture calculations and do not turn those centered correlation summaries into exact admission tests.

The FNO anneal improves its validation RMS error from 5.047% to 4.991%, while its descriptive correlation with the normalized-MSE network increases from 0.95–0.97 to 0.974–0.980. The five-member validation quadratic optimum moves from 4.940% to 4.940%. These measured changes are consistent with an accuracy improvement being offset by greater residual alignment. Proposition 6.7 gives a local sensitivity calculation for fixed positive RMS errors and uncentered alignment. It is not used here as an exact quantitative explanation based on the early centered-correlation summaries.

For an exactly equicorrelated equal-error family, Proposition 6.1(ii) gives the limiting RMS value $\bar{e}\sqrt{\bar{\varrho}}$. Substituting the measured mean RMS error and mean centered correlation gives the older descriptive references $4.922 \pm 0.056\%$ and $4.908 \pm 0.057\%$. The empirical pools

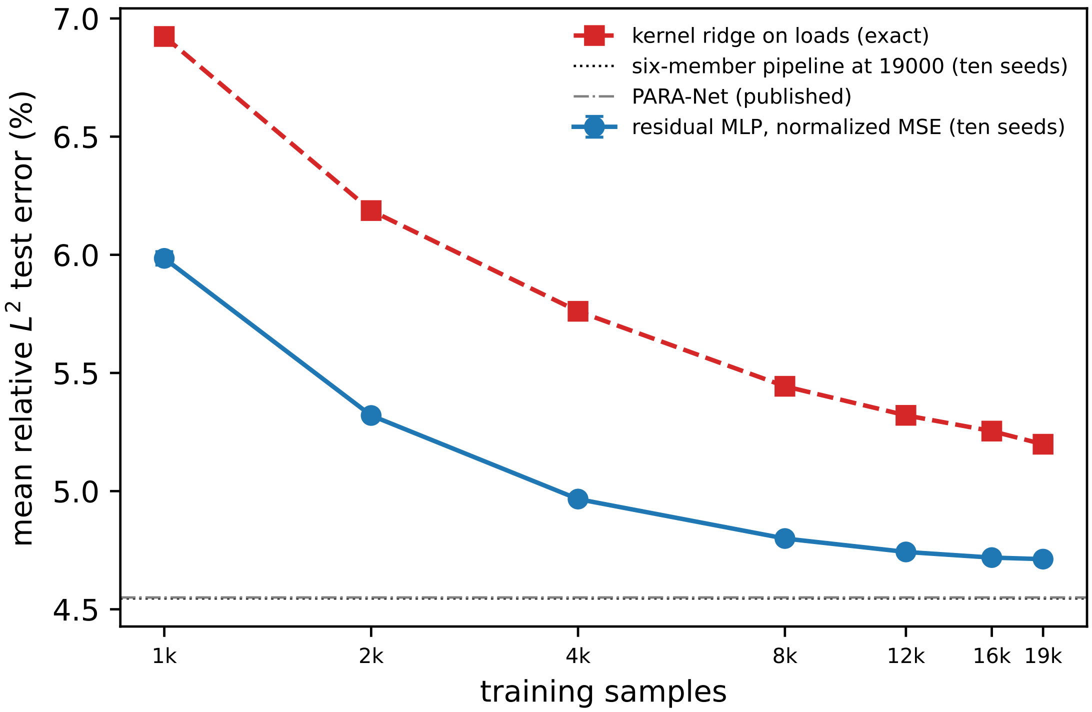

Figure S6.5: Learning curves at ten seeds under one protocol (the first N rows of the training pool, a fixed 1000-case validation carve, the full test block): mean relative test error of the normalized-MSE network, mean and standard deviation over ten seeds, and of the deterministic Matérn kernel ridge on loads, against the number of training samples, both axes logarithmic. The horizontal lines mark the six-member pipeline at 19000 samples and the quoted PARA-Net result. Error bars are smaller than the markers above 4000.

are heterogeneous, so these substitutions are heuristic diagnostics, not lower bounds. The one-sided entrywise bounds below and the sixty-member bound use the actual uncentered second-moment matrices instead.

All these bounds concern RMS error $\mathcal{E}_2$ for global convex weights. Jensen's inequality gives $\mathcal{E}_1 \le \mathcal{E}_2$ for the same predictor, so an RMS lower bound does not transfer to mean error. Its per-pixel affine weights and kernel correction also place it outside the global convex class. Among global sum-to-one weights the signed and convex optima differ by at most 0.00052 percentage points in the stored records, but they do not always coincide: the signed minimizer has a negative coordinate at seven six-member seeds and one five-member seed.

The selected input-space residual correction improves the five-member stack by 0.024 points at $n = 19000$. Figure S6.5 reports the observed response to sample size: the normalized-MSE network goes from $5.985 \pm 0.028\%$ at $N = 1000$ to $4.799 \pm 0.003\%$ at 8000 and $4.712 \pm 0.004\%$ at 19000; the kernel goes from 6.924% to 5.444% and 5.198%. Over the measured range above 8000, fitted log-log slopes are -0.022 ± 0.001 for the network and -0.052 for the kernel. The last step gains 0.006 ± 0.004 points, positive at nine seeds; the last two steps gain 0.030 ± 0.010, positive at all ten. These are finite-range fits, not asymptotic rates. A three-parameter fit $e_\infty + cN^{-\alpha}$ gives different extrapolated asymptotes, 4.64% and 4.88%, so it does not locate a common floor. The observed 0.034-point seed span of our pipeline is also not an estimate of the unreported training variability of published architectures. Shared approximation bias and target discretization artifacts remain unresolved explanations for the residual plateau.

The five-member test-set check of Corollary 6.4 reports a normalized bracket $[0.949 \pm 0.005, 1.039 \pm 0.004]$ and fitted-global-stack quadratic risk 0.980 ± 0.003. That risk uses the stored deployed global weights, not a test-set simplex minimization. The six-member validation-matrix calculation reports a bracket $[0.933 \pm 0.007, 1.032 \pm 0.004]$ and normalized simplex optimum 0.972 ± 0.005. The corollary upper-bounds the minimum by the uniform-weight risk; an arbitrary fitted ensemble need not lie below that upper bound, although the five-member weights do.

From the released six-member validation matrices, the actual one-sided RMS floor $\sqrt{\min_{i \neq j} S_{ij}}$ is $4.820 \pm 0.062\%$, the stronger finite-six-member lower bound is $4.849 \pm 0.060\%$, and the quadratic optimum is $4.920 \pm 0.057\%$. These are different from the heuristic equicorrelation references. The dispersion factors 0.938 ± 0.003 and 0.938 ± 0.003 compare $\mathcal{E}_1$ and $\mathcal{E}_2$ on the same validation data used to compute the weights. They are same-sample metric diagnostics. For six members the ratio of test mean error to validation RMS error is 0.9357 ± 0.0110; that ratio additionally includes transfer between samples and is not a pure dispersion factor.

The sixty-predictor pool. The analysis of Section 4.6 uses all available test predictions of the second campaign (`campaign/seedarch.py`, record `campaign/collected/dgx/seedarch.json`). A fixed permutation divides the test block into 1000 fitting and 19000 evaluation cases for this pooling exercise. Convex weights fitted on the first subset are evaluated on the second; the hindsight optimum is explicitly optimized on the evaluation matrix. Equal-weight seed curves average over all subsets for RMS error; mean errors use all subsets where there are at most 45, and 40 random subsets otherwise.

Using the evaluation matrix, the smallest diagonal is 0.0024512070 and the within- and between-architecture lower second moments divided by that diagonal are 0.9807963 and 0.9377968. Theorem 6.6 then gives 4.814%, below the hindsight RMS optimum 4.876%. This is an algebraic empirical lower bound for global convex combinations of these sixty predictors on these cases. It is not a population lower confidence bound or a restriction on untrained predictors. An extension to more seeds would require the same entrywise lower bounds to hold for those additional members.

Within the existing ten-seed blocks, square roots of the smallest within-architecture second moments are approximately 4.93% for the FNO, 4.96% for the normalized-MSE network, 4.97% for the refiner, 4.94% for the UNet, 4.90% for the metric-loss MLP and 5.46% for the kernel. The corresponding ten-seed equal-weight errors lie close to these empirical references. Replacing lower moments by mean moments in the block formula returns the uniform sixty-member mixture's value exactly by algebra, not by an independent predictive test.

At the pipeline's fixed ridge, per-pixel affine fitting on all sixty members has evaluation error 4.682%, versus 4.556% for the six ten-seed means and 4.567% for six members at

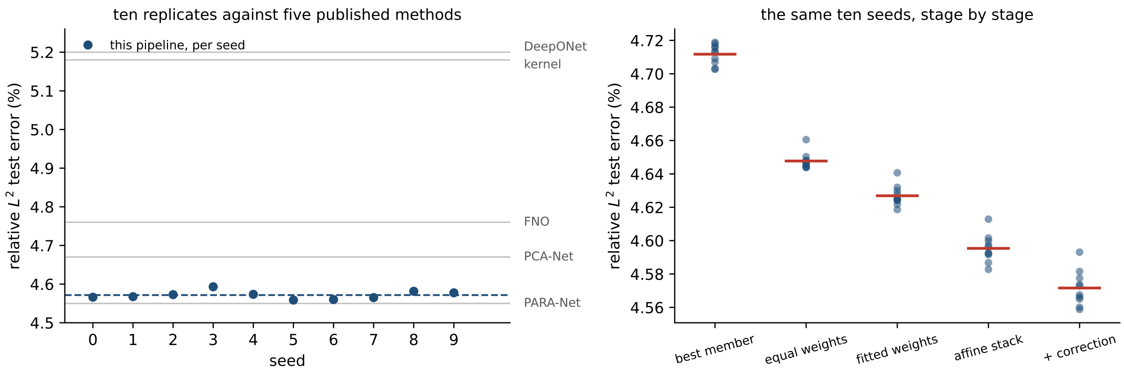

Figure S6.6: Left: the corrected pipeline's test error at each of the ten seeds (points), their mean (dashed), and the five published single numbers on the same axis (grey). These show the variability of our pipeline; comparable seed distributions are unavailable for the published methods. Right: the same ten seeds followed down the pipeline, each stage scored on test, with the stage mean marked. Fitting per-pixel weights lowers the mean; seed spreads at each stage are reported in the text.

seed 0. The leave-one-out ridge comparisons in Section 4.6 show that this difference depends on regularization; it is not evidence that sixty members intrinsically overfit. Removing the UNet seeds changes the uniform-weight mean error from 4.611% to 4.630%.

For the first campaign the test-error seed standard deviation is 0.005 points at uniform weights, 0.006 for fitted global weights, 0.008 for the affine stack and 0.010 after correction. These measurements describe the variability of the fitted stages.

S7 The OCO-2 input metric, and data scaling

S7.1 What the metric buys, measured

The original campaign refits both raw-input kernel rows at five nested training sizes over a sixteenfold range, at ten seeds on each band. Each sensitivity map is estimated from the training rows available at its own sample size. The singular-value counts retaining 99% of the map's squared energy are 12/20, 14/24 and 13/24; the corresponding participation ratios are 11.99, 13.63 and 13.33. These are descriptive summaries of an estimated linear sensitivity map, not identified dimensions of the nonlinear target.

Table S7.1 reports finite-range log–log slopes. On O2 the paired slope difference is small relative to its seed variation. The adapted kernel's fitted exponent is larger on the two CO_2 bands, by 0.011 and 0.052. At $n = 18000$ the isotropic-to-adapted error ratios are 1.353 ± 0.008, 1.118 ± 0.002 and 1.514 ± 0.114. Thus the tested rescaling improves prediction at that training size but leaves much of the raw-input-to-feature-kernel difference. These observations do not establish asymptotic rates. Proposition S2.5 is instead a conditional finite-design rank-truncation bound; the estimated ranks alone do not verify its Lipschitz, domain-radius, or native-space assumptions. All entries here are retained campaign results; no networks or kernels were trained for this comparison.

Table S7.1: The raw-input kernel refit at nested training sizes, ten seeds per band. "Rank" counts singular values of the training-only estimated sensitivity map holding 99% of the squared energy, out of the input dimension. The exponent difference is paired within seed; its standard deviation is over seeds. The last column is the isotropic-to-adapted error ratio at $n = 18000$. The slopes are descriptive fits over the finite sizes tested, not asymptotic convergence rates.

Band	Rank	−slope iso	−slope adapted	difference	ratio at n=18000
O2	12/20	0.221 ± 0.007	0.215 ± 0.017	$+0.006 \pm 0.015$	1.353 ± 0.008
weak CO_2	14/24	0.086 ± 0.005	0.097 ± 0.006	-0.011 ± 0.009	1.118 ± 0.002
strong CO_2	13/24	0.203 ± 0.029	0.255 ± 0.020	-0.052 ± 0.021	1.514 ± 0.114

At its largest size the sweep recovers the raw-kernel rows of Table 6 to the digit, seed by seed and including the selected hyperparameters. We record that as a consistency check on the two code paths and nothing more: both call the same tuning routine on the same standardized inputs, so the agreement is a re-execution of one deterministic computation rather than corroboration by an independent measurement. One protocol note: the table's matched 250-epoch budget is short of convergence for the network rows. A separate single-seed 750-epoch run of the same code — not a row of Table 6, all of whose numbers are at 250 epochs — reaches 3.82% (flat network) and 3.71% (feature head) on this band, with the combination at 3.71% reduced and 0.0174% radiance. This single run shows that the reported levels depend on the training budget.

The training-loss comparison further distinguishes coordinate weighting from per-case weighting. The second network in Table 6 is trained on a diagonally weighted loss over the reduced coefficients, $\|s_z \odot (\hat{z} - z)\|/\|s_z \odot z\|$. That is a surrogate for the radiance-relative error, not the error itself: the reported radiance metric maps predictions through the stored PCA reconstruction and divides by the reconstructed-radiance norm, and the surrogate omits the reconstruction. The third network is trained in the exact radiance metric, reconstruction and denominator included, and it does not win its own metric: it scores *worse* there than the surrogate-trained one, $0.0571 \pm 0.0057\%$ against $0.0442 \pm 0.0044\%$ on O2 and $0.093 \pm 0.008\%$ against $0.075 \pm 0.003\%$ on SCO2, the surrogate ahead at ten of ten seeds on both bands, with WCO2 the reverse at eight of ten ($0.064 \pm 0.015\%$ against $0.072 \pm 0.007\%$). The kernel heads built on their features repeat the pattern. Tripling the training budget narrows the O2 gap (0.0212% against 0.0201% at 750 epochs) without reversing it, but these finite-budget comparisons do not identify why one training loss performs better. By the reconstruction identity, the two losses share the same numerator and differ in their target-dependent denominator, which reweights cases. Both concentrate on the leading coefficients, while their optimization and generalization behavior can differ. The coordinate profile is consistent with this trade-off: the kernel-flow emulator predicts the leading coefficient to 0.0012 root mean square against the flat network's 0.0054, whereas the flat network is three to four times more accurate on the trailing thirty coordinates. These measurements support trying both losses; they do not imply a generally optimal loss.

The flat-network seeds permit a second-moment analysis similar to that on structural mechanics. Proposition 6.1 is stated for the squared-metric risk, so the measurement is made there: for each band we form the matrix S of the ten seeded copies of the flat network on the test set, in the relative metric, and measure the risks of mixtures of that finite pool.

The identity itself holds numerically. At uniform weights $w^\top Sw$ and the directly measured risk of the averaged prediction agree to 7×10^{-18}, 3×10^{-17} and 1×10^{-17} on the three bands — the proposition's content, checked against real predictions rather than assumed.

Over the 45 seed pairs per band, the residual alignments are 0.787 ± 0.010 (O2), 0.971 ± 0.001 (WCO2) and 0.961 ± 0.001 (SCO2). An equicorrelation summary gives ten-member RMS relative errors of 4.419%, 17.997% and 9.267%, matching the recorded uniform-mixture risks at the displayed precision. The corresponding infinite-family values, 4.360%, 17.970% and 9.248%, are extrapolations under the equicorrelation model. They are not lower bounds for untrained seeds or for the architecture class. The exact finite-pool statement is the quadratic identity applied to the measured matrix S.

Comparisons with another readout must use the same metric, since S determines a squared-relative-error quantity and Table 6 reports the mean relative error $\mathbb{E}\,\ell$. Measured in RMS relative error, the Matérn head on learned features reaches 4.575%, 18.11% and 9.326% — better than an individual seeded network (4.915%, 18.238%, 9.433%) and worse than that network's ten-seed ensemble. The ranking is the same in the mean metric, where the seed ensemble reaches 3.916% against the head's 4.111% on O2. At this budget a different kind of member buys less than ensembling ten copies of the plain one; the order-of-magnitude result of this section is the representation ladder, which does not depend on this comparison. These are comparisons of predictors with different computational costs.

Lower residual alignment alone does not guarantee a useful mixture: the weaker member's RMS error also enters Proposition 6.1(i). A original WCO2 Fourier-feature bandwidth sweep records error near 31% with correlation 0.54, and errors above 140% with correlations near 0.11 (Figure S7.1). The surviving log contains rounded errors and correlations, but neither the prediction arrays nor the exact moment convention. It therefore supports an exploratory comparison, not a numerical test of the proposition's admission condition. Adding the exploratory architecture candidates to coordinate selection changed the mean error by less than 0.01 percentage points on each band. This is evidence about this trained pool, not a lower bound for an architecture class.

For the raw-input kernel against the flat network, the stored validation summary has $\varrho = 0.156$ and $e_n/e_k = 0.157$, with RMS relative errors 8.41% and 53.5%. The sign of the criterion changes across seeds, and the optimal mixture gains only a few ten-thousandths of a percentage point. This explains the pair's small contribution.

The combination chooses each coordinate from one of the three networks and three feature-kernel heads using coordinatewise validation RMSE. By Proposition S2.2, this minimizes an auxiliary separable squared loss over that candidate product. Fixed sample denominators retain separability for squared relative losses, with different sample weights for the reduced and radiance criteria. The reported means of relative norms instead couple coordinates through the square root. Consequently this selector is not guaranteed to optimize either reported metric, or both simultaneously. The same selected coordinates produce both reported columns; the baseline emulator's predictions are not candidates. Its seed-averaged errors are lower than those of the rescored kernel-flow emulator in both criteria on all three bands: O2 reaches $4.12 \pm 0.05\%$ against 16.89% reduced and $0.0294 \pm 0.0020\%$ against 0.0448% radiance, SCO2 $8.08 \pm 0.01\%$ against 16.14% and $0.0584 \pm 0.0041\%$ against 0.1147%,

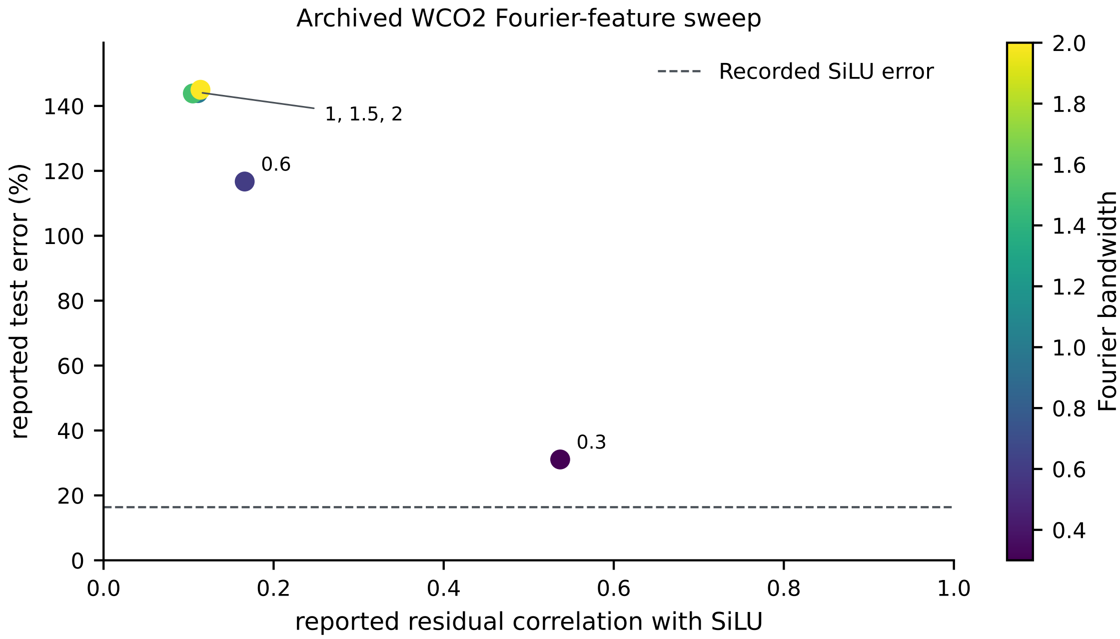

Figure S7.1: Original WCO2 Fourier-feature sweep. Labels and colors identify bandwidths; the dashed line is the log's SiLU error of 16.37%. Three bandwidths nearly coincide in the upper left. The axes report the log's quantities. No theorem boundary is plotted. The source log is `runs/fourier_wco2.log`; its hash and the five rounded pairs are retained in the figure's provenance record.

and WCO2 $16.28 \pm 0.06\%$ against 24.06% and $0.0507 \pm 0.0052\%$ against 0.0599%. Counted by seed, the combination beats the emulator on both metrics at ten of ten seeds on O2 and SCO2 and at nine of ten on WCO2, the single exception missing on the radiance metric by 0.0015 points.

On O2 the combination's reduced mean is slightly worse than that of the flat-feature head (4.11510% versus 4.11080%), while its radiance mean improves from 0.07927% to 0.02940%. The result is therefore a useful trade-off, not dominance over every component. WCO2 has a weaker radiance comparison: one seed is worse than the source emulator by 0.0015 percentage points. The stored paired-bootstrap interval for that seed is $[-0.0012, 0.0042]$ percentage points, while the other nine intervals are negative. These are descriptive intervals on the public test cases, without a correction for repeated benchmark analysis.

S7.2 Learning curves and training budgets

The OCO-2 learning curve decreases more strongly with training size than the structural-mechanics curve over the ranges examined. The left panel of Figure S7.2 shows a one-seed O2 experiment with a disjoint held-out set. A finite-range log–log fit gives slope -0.68, with error decreasing from about 74% at a few hundred pairs to 4.3% at $n = 17700$. The main ten-seed combination at 18000 training pairs has mean 4.12% at its 250-epoch budget; the 3.71% result above belongs to a separate single-seed 750-epoch experiment. These are different protocols and should not be combined into one scaling fit. The raw-input kernel improves more slowly over the sizes. A separate full-resolution $58 \to 3048$ experiment shows little change over a few hundred retrievals. Neither observation identifies the limiting source of error or an asymptotic convergence rate.

The retained ClimSim records cover a larger data range. ClimSim (Yu et al., 2023) is a climate emulation dataset of about ten million samples that maps a 124-dimensional atmospheric state to 128 physics tendencies, with published neural baselines. The runs compare two model families at one, two and three and a half million training samples under the recorded protocol, with within-run selection on a held-out block drawn from the training pool and separate evaluation data; the right panel of Figure S7.2 plots those points. Each campaign seed draws its own 20000-row test block, so the spreads below include that variation.

What the points establish is the high-data half of the picture. The neural mean has higher R^2 than the kernel and increases with size: 0.544 ± 0.005 at one million (five seeds), 0.563 ± 0.004 at two million and 0.575 ± 0.004 at three and a half million (three seeds each), compared for orientation with a published multilayer-perceptron value near 0.6; a matched split-and-scoring reproduction of that baseline is outside the scope of this comparison. The kernel rows of the campaign read 0.390 ± 0.007, 0.391 ± 0.006 and 0.387 ± 0.007 at the three sizes, but that flatness is by construction and says nothing about the kernel: the exact solve was fitted on a 6000-row subsample of the training slice at every size, a budget rather than a data point. What the kernel does with more rows was measured afterwards under the same protocol (selection on the pool validation block, separate test evaluation, seed 0, the one-million slice): 0.389 at 6000 rows, 0.420 at 12000 and 0.432 at 24000 (0.446 at a second seed, whose test block is a different draw). It improves with budget, by about three hundredths for the first doubling and one for the second, and at 24000 rows — four times the campaign's budget, and the largest exact solve we ran — it remains eleven hundredths of R^2 below the network at one million samples. The comparison remains unequal in training size and total computation: the neural mean uses up to millions of examples, whereas the exact kernel uses at most 24000. The records show an advantage for the network under those budgets; they do not establish which method would win at matched compute or larger kernel budgets. The smaller-n ClimSim results from an earlier analysis were withdrawn because of leakage and are not evidence here. The retained range does not locate a crossover between model families.

S8 Proofs

S8.1 Proof of Proposition S1.1

By convexity of $\ell(\cdot, v)$,

$$\ell\big(\widehat{G}_S(u), G(u)\big) \;\le\; \tfrac{1}{2}\,\ell\big(\widehat{G}(u), G(u)\big) + \tfrac{1}{2}\,\ell\big(T\widehat{G}(Su), G(u)\big).$$

Since T is an involution, the equivariance $G(Su) = TG(u)$ applied at u gives $TG(Su) = T^2G(u) = G(u)$, hence

$$\ell\big(T\widehat{G}(Su), G(u)\big) = \ell\big(T\widehat{G}(Su), TG(Su)\big) = \ell\big(\widehat{G}(Su), G(Su)\big),$$

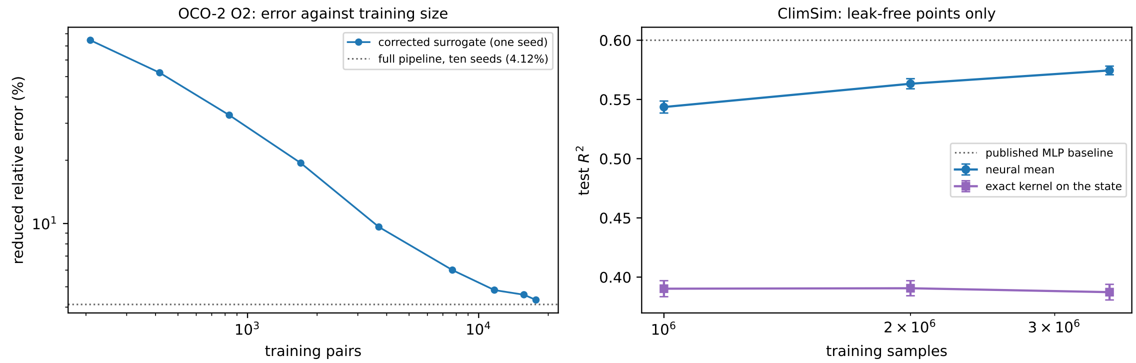

Figure S7.2: Error against training-set size, drawn by `campaign/make_scaling_fig.py`. Left: the OCO-2 O2 task, reduced error of the corrected surrogate at one seed, from the data-scaling run; a finite-range log–log fit has slope -0.68, and the recorded error is 4.3% at $n = 17700$, against the 4.12% the ten-seed pipeline reaches on all 18000 pairs (dotted). Right: ClimSim, test R^2 over the outputs with non-negligible variance, at one, two and three and a half million training samples under the held-out selection protocol — five seeds at one million and three at each of the larger sizes, error bars over seeds. No crossover between the two families is observed over this range. The kernel points are the campaign's, fitted on 6000 rows at every training size, so their flatness across sizes is by construction; the kernel's own budget ladder, measured afterwards under the same protocol, is in the text. The neural mean has higher R^2; the published baseline (dotted) is an orientation point rather than a matched-protocol comparison.

using the T-invariance of the loss. Taking expectations and using the S-invariance of μ (so that $Su \sim \mu$ when $u \sim \mu$),

$$\mathbb{E}\,\ell\big(\widehat{G}_S(u), G(u)\big) \le \tfrac{1}{2}\,\mathbb{E}\,\ell\big(\widehat{G}(u), G(u)\big) + \tfrac{1}{2}\,\mathbb{E}\,\ell\big(\widehat{G}(Su), G(Su)\big) = \mathbb{E}\,\ell\big(\widehat{G}(u), G(u)\big). \quad \square$$

S8.2 Proof of Proposition S1.2

Strict positive definiteness of the restricted kernel matrix makes I_C the (unique) interpolant of the pairs indexed by C, so $I_C(z_j) = v_j$ for $j \in C$ and the terms of (S1.2) indexed by C vanish, which is the first claim. For the second, write

$$e_2(\theta; C) \;=\; \sum_{i \in B \setminus C} g(C, i), \qquad g(C, i) := \|v_i - I_C(z_i)\|_2^2.$$

Conditionally on C, the sum has exactly $b/2$ terms, so $e_2(\theta; C) = \frac{b}{2}\,\mathbb{E}_{i \sim \mathrm{Unif}(B \setminus C)}[g(C, i)]$, and taking the expectation over C proves the identity. The gradient claim is immediate: for fixed (B, C) the map $\theta \mapsto e_2(\theta; C)$ is differentiable wherever the Cholesky factorization in I_C is (the kernel matrix stays positive definite in a neighborhood), and under a local integrable Lipschitz bound the derivative passes under the expectation, so $\mathbb{E}_{B,C}[\nabla_\theta e_2] = \nabla_\theta\,\mathbb{E}_{B,C}[e_2]$. $\square$

S8.3 Proof of Proposition S1.3

(i) Since $\sum_m w_m = 1$, $F_w(u) - G(u) = \sum_m w_m\,(f_m(u) - G(u))$, and the triangle inequality gives $\|F_w(u) - G(u)\|_2 \le \sum_m w_m \|f_m(u) - G(u)\|_2$. Dividing by $\|G(u)\|_2$ yields the pointwise claim; taking expectations gives the risk inequality, and bounding each $\ell(f_m(u), G(u))$ by B gives $\ell(F_w(u), G(u)) \le B$.

(ii) For $M = 1$ there is no weight selection and the asserted inequality is immediate. Suppose $M \ge 2$ and write $\ell_u(w) = \ell(F_w(u), G(u))$. First, ℓ_u is Lipschitz on Δ for the ℓ^1 norm: for $w, w' \in \Delta$, the reverse triangle inequality and the argument of (i) give

$$|\ell_u(w) - \ell_u(w')| \;\le\; \frac{\big\|\sum_m (w_m - w'_m)(f_m(u) - G(u))\big\|_2}{\|G(u)\|_2} \;\le\; B\,\|w - w'\|_1.$$

Next, discretize the simplex. For $k \in \mathbb{N}$ let $\mathcal{G}_k = \{w \in \Delta : \, kw \in \mathbb{Z}^M\}$; its cardinality is the number of compositions of k into M nonnegative parts, $\binom{k+M-1}{M-1} \le (k+1)^{M-1}$. Given $w \in \Delta$, set $w'_i = \lfloor k w_i \rfloor / k$ for $i < M$ and $w'_M = 1 - \sum_{i<M} w'_i$; then $w' \in \mathcal{G}_k$ (the last coordinate is a multiple of $1/k$ and is $\ge w_M \ge 0$ because the first $M-1$ coordinates only decreased), and

$$\|w - w'\|_1 = \sum_{i<M} (w_i - w'_i) + \big(w'_M - w_M\big) = 2\sum_{i<M} (w_i - w'_i) \;\le\; \frac{2(M-1)}{k}.$$

By (i) the per-sample loss lies in $[0, B]$, so for each fixed w' Hoeffding's inequality gives $\mathbb{P}\big(|\widehat{R}(w') - R(w')| > t\big) \le 2e^{-2mt^2/B^2}$, and a union bound over $\mathcal{G}_k$ shows that with probability at least $1-\delta$,

$$\sup_{w' \in \mathcal{G}_k} \big|\widehat{R}(w') - R(w')\big| \;\le\; B\sqrt{\frac{\log\big(2|\mathcal{G}_k|/\delta\big)}{2m}}.$$

On this event, for every $w \in \Delta$, approximating by its grid point and using the Lipschitz property for both R and $\widehat{R}$,

$$\big|\widehat{R}(w) - R(w)\big| \;\le\; B\sqrt{\frac{\log(2|\mathcal{G}_k|/\delta)}{2m}} + \frac{4B(M-1)}{k}.$$

If $w^\star$ minimizes R over Δ, the empirical minimizer satisfies $R(F_{\widehat{w}}) \le \widehat{R}(\widehat{w}) + \sup_\Delta |\widehat{R} - R| \le \widehat{R}(w^\star) + \sup_\Delta |\widehat{R} - R| \le R(F_{w^\star}) + 2\sup_\Delta |\widehat{R} - R|$. Choosing $k = m(M-1)$ and using $|\mathcal{G}_k| \le (m(M-1)+1)^{M-1}$ gives

$$R(F_{\widehat{w}}) \;\le\; \min_{w\in\Delta} R(F_w) + \frac{8B}{m} + B\sqrt{\frac{2\big((M-1)\log(m(M-1)+1) + \log(2/\delta)\big)}{m}},$$

which is the claim. $\square$

S8.4 Proof of Lemma S1.4

For each k the validation scores $\ell(g_k(\tilde{u}_i), G(\tilde{u}_i))$, $i \le m'$, are i.i.d., take values in $[0, B]$, and have mean $R(g_k)$; the maps g_k are fixed relative to this sample, which is the only place the independence assumption enters. Hoeffding's inequality gives, for each k,

$$\Pr\Big(\big|\widehat{R}(g_k) - R(g_k)\big| \ge t\Big) \;\le\; 2\exp\big(-2m't^2/B^2\big),$$

so with $t = B\sqrt{\log(2K/\delta)/(2m')}$ and a union bound over $k \le K$, all K deviations are below t simultaneously with probability at least $1-\delta$. On that event, if k^* attains $\min_k R(g_k)$,

$$R(g_{\widehat{k}}) \;\le\; \widehat{R}(g_{\widehat{k}}) + t \;\le\; \widehat{R}(g_{k^*}) + t \;\le\; R(g_{k^*}) + 2t,$$

and $2t = B\sqrt{2\log(2K/\delta)/m'}$. □

S8.5 Proof of Proposition S1.5

Fix a coordinate j and abbreviate $\varphi = \varphi_j$, $Y(u) = G(u)_j$, $D = \sqrt{M+1}$. Each entry of $\varphi(u)$ is bounded by b in absolute value (the members by assumption, the intercept entry by construction), so $\|\varphi(u)\|_2 \le bD$, and for $\|w\|_2 \le W$ Cauchy–Schwarz gives $|w^\top\varphi(u)| \le WbD$ and $|w^\top\varphi(u) - Y(u)| \le b(1+WD)$. The losses $\ell_w(u) = (w^\top\varphi(u) - Y(u))^2$ therefore take values in $[0, L_\infty]$ with $L_\infty = b^2(1+WD)^2$.

Uniform deviation. Consider the one-sided suprema $Z^+ = \sup_{\|w\|_2\le W}\big(\widehat{R}_j(w) - R_j(w)\big)$ and $Z^- = \sup_{\|w\|_2\le W}\big(R_j(w) - \widehat{R}_j(w)\big)$ over the validation sample of size m. Changing one sample moves either supremum by at most L_∞/m, so McDiarmid's inequality gives $Z^\pm \le \mathbb{E}Z^\pm + L_\infty\sqrt{\log(4q/\delta)/(2m)}$, each with probability at least $1-\delta/(2q)$. By symmetrization, $\mathbb{E}Z^\pm \le 2\,\mathfrak{R}_m(\mathcal{L})$, the Rademacher complexity of the loss class. The loss is the composition of $t \mapsto t^2$, restricted to $|t| \le b(1+WD)$ where it is $2b(1+WD)$-Lipschitz, with the class $\{u \mapsto w^\top\varphi(u) - Y(u)\}$; translation by the fixed function Y leaves the Rademacher average unchanged (the translated term does not involve w and has zero mean), and for the linear class over the ℓ^2 ball,

$$\mathfrak{R}_m \;\le\; \frac{W}{m}\,\mathbb{E}\Big\|\sum_{i=1}^m \sigma_i\varphi(\tilde{u}_i)\Big\|_2 \;\le\; \frac{W}{m}\Big(\sum_{i=1}^m \mathbb{E}\|\varphi(\tilde{u}_i)\|_2^2\Big)^{1/2} \;\le\; \frac{WbD}{\sqrt{m}},$$

by Jensen's inequality. Talagrand's contraction lemma then gives $\mathfrak{R}_m(\mathcal{L}) \le 2b(1+WD)\cdot WbD/\sqrt{m}$, hence

$$Z^\pm \;\le\; \frac{4Wb^2D(1+WD)}{\sqrt{m}} + L_\infty\sqrt{\frac{\log(4q/\delta)}{2m}} \qquad \text{jointly with probability} \ge 1-\delta/q.$$

Empirical risk minimization. On this event, for any minimizer w^* of R_j over the ball, $R_j(\widehat{w}_j) \le \widehat{R}_j(\widehat{w}_j) + Z^- \le \widehat{R}_j(w^*) + Z^- \le R_j(w^*) + Z^+ + Z^-$, and $Z^+ + Z^-$ is bounded by the displayed excess. A union bound over the two tails of each of the q coordinates makes all events hold simultaneously with probability at least $1-\delta$, and averaging over j preserves the (coordinate-uniform) bound. This proves (i).

For (ii), the ridge minimizer satisfies $\widehat{R}_j(\widehat{w}_j) + \lambda\|\widehat{w}_j\|_2^2 \le \widehat{R}_j(w^*) + \lambda\|w^*\|_2^2$, so $\widehat{R}_j(\widehat{w}_j) \le \widehat{R}_j(w^*) + \lambda\|w^*\|_2^2 \le \widehat{R}_j(w^*) + \lambda W^2$. Since by assumption $\|\widehat{w}_j\|_2 \le W$, the uniform event of part (i) applies to $\widehat{w}_j$ and w^* alike, and the chain above goes through with the extra λW^2. □

S8.6 Proof of Corollary 6.4

The lower bound is Corollary 6.3. For the upper bound, evaluate the quadratic form at the uniform weights $w_m = 1/M$:

$$\frac{1}{\bar{e}^2}\, w^\top S w = \frac{1}{\bar{e}^2}\Big(\frac{1}{M^2}\sum_m S_{mm} + \frac{1}{M^2}\sum_{m\neq k} S_{mk}\Big) \;\le\; \frac{1+\delta_e}{M} + \frac{M-1}{M}\,\big(\bar{\varrho}+\delta_\varrho\big),$$

using the entrywise upper bounds and the counts M and $M(M-1)$ of the two sums. The right side equals $\bar{\varrho}+\delta_\varrho + \big(1+\delta_e - \bar{\varrho} - \delta_\varrho\big)/M$, and the minimum over the simplex is at most the value at any feasible point. $\square$

S8.7 Proof of Corollary S1.6

Fix a coordinate j and a grid point γ. Substituting the affine form of the correction,

$$h_{w,\gamma}(u)_j = w_j^\top \widetilde{\varphi}_{j,\gamma}(u) + o_{j,\gamma}(u), \qquad \widetilde{\varphi}_{j,\gamma}(u) = \varphi_j(u) - \Phi_j^\top c_\gamma(u), \quad o_{j,\gamma}(u) = c_\gamma(u)^\top Y_j,$$

and neither $\widetilde{\varphi}_{j,\gamma}$ nor $o_{j,\gamma}$ depends on w or on the validation sample: c_γ is built from the training design alone. Each component of $\Phi_j^\top c_\gamma(u)$ is a c_γ-weighted sum of member values bounded by b, so its absolute value is at most $b\,\|c_\gamma(u)\|_1 \le b\kappa$, whence $\|\widetilde{\varphi}_{j,\gamma}(u)\|_2 \le bD + bD\kappa = bD(1+\kappa)$ and $|o_{j,\gamma}(u)| \le b\kappa$. For $\|w\|_2 \le W$ the prediction deviates from the target by at most $WbD(1+\kappa) + b\kappa + b \le b(1+WD)(1+\kappa) =: \tilde{c}$, so the squared losses lie in $[0,\tilde{c}^{\,2}]$ and are $2\tilde{c}$-Lipschitz in the prediction.

The argument of Proposition S1.5 now applies verbatim to the class indexed by w in the W-ball, for this fixed (j,γ): the Rademacher complexity of the affine part is at most $W\,bD(1+\kappa)/\sqrt{m}$, contraction multiplies it by $2\tilde{c}$, symmetrization doubles it, and McDiarmid at level $\delta/(2q|\Gamma|)$ adds $\tilde{c}^{\,2}\sqrt{\log(4q|\Gamma|/\delta)/(2m)}$. A union bound over the two tails of the $q|\Gamma|$ pairs makes, with probability at least $1-\delta$, each one-sided deviation $Z^{\pm}_{j,\gamma}$, defined as in the proof of Proposition S1.5, at most

$$\frac{4Wb^2D(1+WD)(1+\kappa)^2}{\sqrt{m}} + b^2(1+WD)^2(1+\kappa)^2\sqrt{\frac{\log(4q|\Gamma|/\delta)}{2m}}.$$

On that event, write $\bar{R}(\gamma) = \frac{1}{q}\sum_j R_j(h_{\widehat{w},\gamma})$ and $\widehat{\bar{R}}(\gamma)$ for its empirical version; averaging the $Z_{j,\gamma}$ over j bounds $|\widehat{\bar{R}}(\gamma) - \bar{R}(\gamma)|$ by the same quantity for every γ simultaneously, including at the data-dependent $\widehat{w}$, because the deviations are uniform over the ball. The selection chain $\bar{R}(\widehat{\gamma}) \le \widehat{\bar{R}}(\widehat{\gamma}) + Z \le \widehat{\bar{R}}(\gamma^*) + Z \le \bar{R}(\gamma^*) + 2Z$ with γ^* the aggregate oracle then gives the display, whose constant is $2Z$ in the factored form stated. When the c_γ are kernel ridge weights, Lemma S1.7 supplies the hypothesis with $\kappa = \kappa_{\mathrm{an}}$. $\square$

S8.8 Proof of Lemma S1.7

Fix u and write $k_u = k(X,u)$, $t = n\lambda$, $A = K + tI$ and $c = A^{-1}k_u$. Because k is positive definite, the Gram matrix of the $n+1$ points $u_1,\dots,u_n,u$ is positive semidefinite, and by the generalized Schur complement this places k_u in the range of K with $k_u^\top K^+ k_u \le k(u,u)$,

K^+ the pseudo-inverse. Diagonalize $K = \sum_i \mu_i v_i v_i^\top$ and expand $k_u = \sum_i \beta_i v_i$ over the eigenvectors with $\mu_i > 0$; the Schur condition reads $\sum_i \beta_i^2/\mu_i \le k(u,u)$. Then

$$\|c\|_2^2 = \sum_i \frac{\beta_i^2}{(\mu_i+t)^2} = \sum_i \frac{\beta_i^2}{\mu_i} \cdot \frac{\mu_i}{(\mu_i+t)^2} \;\le\; \frac{1}{4t} \sum_i \frac{\beta_i^2}{\mu_i} \;\le\; \frac{k(u,u)}{4t},$$

because $(\mu+t)^2 \ge 4\mu t$ for all $\mu, t \ge 0$. Cauchy–Schwarz gives $\|c\|_1 \le \sqrt{n}\,\|c\|_2$, and combining, $\|c\|_1^2 \le n\|c\|_2^2 \le k(u,u)/(4\lambda)$.

For the sharpness take $k(x,y) = \cos(x-y)$, a positive definite kernel on the line (its native space is spanned by cos and sin), the two points $x_1 = 0$ and $x_2 = \theta$ with $\theta \in (0,\pi)$ chosen so that $1 - \cos\theta = 2\lambda$, and $u = (\theta+\pi)/2$. Then $K = \left(\begin{smallmatrix} 1 & \cos\theta \\ \cos\theta & 1 \end{smallmatrix}\right)$ has the eigenvector $v = (1,-1)/\sqrt{2}$ with eigenvalue $\mu = 1 - \cos\theta = t$, and $k_u = (\cos u, \cos(u-\theta)) = -\sin(\theta/2)\,(1,-1)$ is proportional to v, with $\beta^2/\mu = 2\sin^2(\theta/2)/(1-\cos\theta) = 1 = k(u,u)$. Hence $c = A^{-1}k_u = k_u/(2t)$ has entries of equal magnitude, $\|c\|_2^2 = \beta^2/(2t)^2 = 1/(4t)$, and $\|c\|_1 = \sqrt{2}\,\|c\|_2 = \sqrt{n}\,\|c\|_2 = \frac{1}{2}\lambda^{-1/2}$, so every inequality in the display is an equality.

For the Matérn kernel of Section 3.3, $k(u,u) = 1$ and the smallest nugget on the grid is 10^{-6}, so $\sup_{u,\gamma} \|c_\gamma(u)\|_1 \le \frac{1}{2}\max_\gamma \lambda_\gamma^{-1/2} = 500$ irrespective of the length scale and of the design. $\square$

S8.9 Proof of Corollary 6.5

The objective is strictly convex on the hyperplane $\{\mathbf{1}^\top w = 1\}$, so its minimizer there is the unique stationary point of the Lagrangian $w^\top S w - \nu(\mathbf{1}^\top w - 1)$, namely $w = \frac{\nu}{2} S^{-1}\mathbf{1}$ with ν fixed by the constraint: $w^\star = S^{-1}\mathbf{1}/(\mathbf{1}^\top S^{-1}\mathbf{1})$, at which $w^{\star\top} S w^\star = 1/(\mathbf{1}^\top S^{-1}\mathbf{1})$. The simplex is a subset of the hyperplane, so the convex minimum is at least this value, and by uniqueness of the minimizer the two agree exactly when $w^\star$ lies in the simplex, that is, has no negative coordinate. $\square$

S8.10 Proof of Proposition 6.1

Since $\sum_m w_m = 1$, the stack's normalized residual is $\rho_{f_w}(u) = \sum_m w_m \rho_{f_m}(u)$, hence

$$\mathbb{E}\,\ell\big(f_w(u), G(u)\big)^2 = \mathbb{E}\Big\|\sum_m w_m \rho_{f_m}(u)\Big\|_2^2 = \sum_{m,k} w_m w_k\,\mathbb{E}\,\langle \rho_{f_m}(u), \rho_{f_k}(u)\rangle = w^\top S w.$$

For (i), parameterize $w = (1-t, t)$; $q(t) = w^\top S w$ is a convex quadratic in t with $q'(0) = 2(\varrho e_1 e_2 - e_1^2)$, negative precisely when $\varrho < e_1/e_2$. In that case the unconstrained minimizer lies in $(0,1)$ and gives the stated interior value; otherwise $q'(0) \ge 0$, the minimum over $[0,1]$ is at $t = 0$ with value e_1^2, and the second member is dropped. For (ii), $S = e^2\big((1-\varrho)I + \varrho\,\mathbf{1}\mathbf{1}^\top\big)$, so $w^\top S w = e^2\big((1-\varrho)\|w\|_2^2 + \varrho\big)$ on the simplex, minimized by the uniform weights, where $\|w\|_2^2 = 1/M$. $\square$

S8.11 Proof of Corollary 6.3

Convex weights are nonnegative, so each term of $w^\top S w = \sum_m w_m^2 S_{mm} + \sum_{m\ne k} w_m w_k S_{mk}$ can be bounded below entrywise:

$$w^\top S w \;\ge\; \bar e^{\,2} \sum_m w_m^2 + \bar\varrho\,\bar e^{\,2} \sum_{m\ne k} w_m w_k = \bar e^{\,2}\Big(\|w\|_2^2 + \bar\varrho\,(1 - \|w\|_2^2)\Big),$$

using $\sum_{m\neq k} w_m w_k = (\sum_m w_m)^2 - \|w\|_2^2 = 1 - \|w\|_2^2$. The bracket is a convex combination of 1 and $\bar\varrho$, hence at least $\bar\varrho$. $\square$

S8.12 Proof of Theorem 6.6

Write $W_a = \sum_k w_{a,k}$ for the total weight of architecture a, so that $\sum_a W_a = 1$. Split the quadratic form into its diagonal terms, the terms between two seeds of one architecture and the terms between architectures, and bound each entry below by its hypothesis; the weights are nonnegative, so the inequalities survive the multiplication:

$$w^\top S w \;\geq\; \bar e^{\,2}\Big(\|w\|_2^2 + \varrho_w\big(\textstyle\sum_a W_a^2 - \|w\|_2^2\big) + \varrho_b\big(1 - \sum_a W_a^2\big)\Big),$$

using $\sum_{k\neq l} w_{a,k}w_{a,l} = W_a^2 - \sum_k w_{a,k}^2$ within an architecture and $\sum_{a\neq b} W_a W_b = 1 - \sum_a W_a^2$ across them. Rearranged, the right side is $\bar e^{\,2}\big(\varrho_b + (\varrho_w - \varrho_b)\sum_a W_a^2 + (1-\varrho_w)\|w\|_2^2\big)$. Both coefficients are nonnegative by hypothesis; $\sum_a W_a^2 \geq 1/M$ by Cauchy–Schwarz, with equality at $W_a = 1/M$; and $\|w\|_2^2 \geq 1/(MK)$ with equality at uniform weights, which also give $W_a = 1/M$. So the minimum of the right side over the simplex is attained at uniform weights and equals the display. When the three hypotheses hold with equality, $S = \bar e^{\,2}\big((1-\varrho_w)I + (\varrho_w - \varrho_b)B + \varrho_b \mathbf{1}\mathbf{1}^\top\big)$ with B the indicator of a shared architecture, the first inequality is an identity for every w, and uniform weights attain the bound. The three limits follow by letting K, then M, grow. $\square$

S8.13 Proof of Proposition 6.7

Proposition 6.1(i) gives $V = e_1^2 e_2^2(1-\varrho^2)/(e_1^2 + e_2^2 - 2\varrho e_1 e_2) = e_1^2\, t^2(1-\varrho^2)/D$ with $D = 1 + t^2 - 2\varrho t$, valid whenever the unconstrained minimizer lies in the open simplex, which is the condition $\varrho < 1/t$ for $t = e_2/e_1 \geq 1$. The strict derivative claims additionally assume $-1 < \varrho$, as in the proposition. Differentiating,

$$\begin{aligned}\frac{\partial V}{\partial t} &= \frac{e_1^2\big[2t(1-\varrho^2)D - t^2(1-\varrho^2)(2t - 2\varrho)\big]}{D^2}\\ &= \frac{2e_1^2\, t(1-\varrho^2)\big(D - t^2 + \varrho t\big)}{D^2}\\ &= \frac{2e_1^2\, t(1-\varrho^2)(1-\varrho t)}{D^2},\end{aligned}$$

and

$$\begin{aligned}\frac{\partial V}{\partial \varrho} &= \frac{e_1^2\big[-2\varrho t^2 D + 2t^3(1-\varrho^2)\big]}{D^2}\\ &= \frac{2e_1^2\, t^2\big(t - \varrho - \varrho t^2 + \varrho^2 t\big)}{D^2}\\ &= \frac{2e_1^2\, t^2(t-\varrho)(1-\varrho t)}{D^2}.\end{aligned}$$

Both are positive for $-1 < \varrho < 1/t$, since $1-\varrho^2 > 0$, $t - \varrho > 0$ and $1 - \varrho t > 0$ there. Setting $\mathrm{d}V = (\partial V/\partial t)\,\mathrm{d}t + (\partial V/\partial \varrho)\,\mathrm{d}\varrho = 0$ and cancelling the common factor $2e_1^2 t(1-\varrho t)/D^2$ gives $(1-\varrho^2)\,\mathrm{d}t + t(t-\varrho)\,\mathrm{d}\varrho = 0$, which is the stated rate; substituting $\mathrm{d}t = -\epsilon t$ gives the fractional form. $\square$

S8.14 Proof of Proposition S2.2

The metric is diagonal, so the squared norm splits over coordinates and the j-th summand $w_j^2\,\mathbb{E}\big(\sum_m v_{mj} f_m(u)_j - G(u)_j\big)^2$ depends on the weights only through the column $v_{\cdot j}$. Minimizing the sum is therefore q independent minimizations, and the positive factor w_j^2 does not move any of the q argmins, so the optimizer is the same for every positive w. A combination with shared weights is the special case $v_{mj} = v_m$ for all j, a subset of the feasible set of each coordinate problem, which gives the domination. $\square$

S8.15 Proof of Corollary S2.3

The integrand is nonnegative, so Tonelli's theorem permits exchanging the expectation with the coordinate sum:

$$\mathbb{E}\Big[a(u)\sum_j w_j^2\big(f_v(u)_j - G(u)_j\big)^2\Big] = \sum_j w_j^2\,\mathbb{E}\Big[a(u)\big(f_v(u)_j - G(u)_j\big)^2\Big].$$

The j-th summand depends on v only through its j-th column $v_{\cdot j}$, exactly as in the proof of Proposition S2.2, so minimizing over v splits into q independent problems; the factors w_j^2 rescale the summands without moving any per-coordinate minimizer, which gives the simultaneous optimality over all diagonal w. The choice $a(u) = \|G(u)\|_2^{-2}$, $w = \mathbf{1}$ identifies the objective with $\mathbb{E}\,\|\rho_{f_v}\|_2^2$, the mean squared relative error. Finally $\mathbb{E}\|\rho\|_2 \le (\mathbb{E}\|\rho\|_2^2)^{1/2}$ is Jensen's inequality. $\square$

S8.16 Proof of Proposition S2.4

Fix a coordinate j and a member m. The validation scores $a(\tilde{u}_i)(f_m(\tilde{u}_i)_j - G(\tilde{u}_i)_j)^2$, $i \le m_v$, are i.i.d., take values in $[0, L]$ by assumption, and have mean $R_j(m)$; the members and the weight are fixed relative to the validation sample. Hoeffding's inequality gives

$$\Pr\Big(\big|\widehat{R}_j(m) - R_j(m)\big| \ge t\Big) \;\le\; 2\exp\big(-2m_v t^2/L^2\big)$$

for each of the Mq pairs (m, j). With $t = L\sqrt{\log(2Mq/\delta)/(2m_v)}$ and a union bound, all Mq deviations are below t simultaneously with probability at least $1-\delta$. On that event, for every j, writing $m^*(j)$ for a population minimizer,

$$R_j\big(\widehat{m}(j)\big) \le \widehat{R}_j\big(\widehat{m}(j)\big) + t \le \widehat{R}_j\big(m^*(j)\big) + t \le R_j\big(m^*(j)\big) + 2t.$$

The metric statement follows by multiplying the coordinate-wise inequality by $w_j^2 \ge 0$ and summing; the selection $\widehat{m}(\cdot)$ does not depend on w, so one event serves every diagonal metric. $\square$

S8.17 Proof of Proposition 6.8

Let $T : \mathcal{H}_k \to \mathbb{R}^{\mathcal{X}}$ be the composition map $Tf = f \circ \varphi$. For $x \in \mathcal{X}$ the reproducing property gives $(Tf)(x) = f(\varphi(x)) = \langle f, k(\cdot, \varphi(x))\rangle_{\mathcal{H}_k}$, so evaluation of Tf at x is a bounded functional. The kernel of T is the intersection of the kernels of these bounded evaluations and is closed. Its orthogonal complement therefore contains a unique minimum-norm preimage of every element in the image. Thus the image $\mathcal{H}_{k_\varphi} := T(\mathcal{H}_k)$ carries the quotient norm $\|g\|_{\mathcal{H}_{k_\varphi}} = \min\{\|f\|_{\mathcal{H}_k} : Tf = g\}$, the minimum over the affine subspace $T^{-1}(g)$ (nonempty exactly when g is a pullback). The quotient by this closed subspace is complete. It is an RKHS, with reproducing kernel $k_\varphi(x, x') = \langle k(\cdot, \varphi(x)), k(\cdot, \varphi(x'))\rangle_{\mathcal{H}_k} = k(\varphi(x), \varphi(x'))$, since the minimum-norm representer of evaluation at x is $k(\cdot, \varphi(x))$. Taking $f = h$ in the minimum gives $\|h \circ \varphi\|_{\mathcal{H}_{k_\varphi}} \le \|h\|_{\mathcal{H}_k}$, and substituting this bound into Theorem 6.9 applied with the kernel k_φ replaces $\|G\|$ by $\|h\|_{\mathcal{H}_k}$. □

S8.18 Proof of Proposition S2.5

The truncated singular-value decomposition satisfies $\|A - A_r\|_{\mathrm{op}} = \sigma_{r+1}$. Consequently, for every $u \in \mathcal{U}$,

$$\|G(u) - G_r(u)\|_2 = \|g(Au) - g(A_r u)\|_2 \le L\|(A - A_r)u\|_2 \le LR\sigma_{r+1} = \varepsilon.$$

Write $\delta(u) = G(u) - G_r(u)$, and form the kernel predictor of G_r using the same feature-space weights $c = c_\lambda(u)$. Theorem 6.9 applied in feature space with zero mean gives

$$\|G_r(u) - c^\top G_r(X)\|_2 = \|h(\varphi_r(u)) - c^\top h(Z)\|_2 \le \|h\|_K\, \widetilde{P}_\lambda(u).$$

The estimator in the proposition is trained on $G(X)$, so

$$G(u) - c^\top G(X) = G_r(u) - c^\top G_r(X) + \delta(u) - \sum_i c_i \delta(u_i).$$

Taking Euclidean norms and using $\|\delta(u_i)\|_2 \le \varepsilon$ yields the stated bound. The argument is deterministic and applies to any design and any stated nugget, including the pseudoinverse convention at zero. □

S8.19 Proof of Theorem 6.9

Fix u and write $k_u = k(X, u) \in \mathbb{R}^n$. At $\lambda = 0$, interpret the inverse as K^+ if K is singular. For $z \in \ker K$, the RKHS function $h_z = \sum_i z_i k(u_i, \cdot)$ has squared norm $z^\top K z = 0$. Hence $z^\top k_u = \langle h_z, k(u, \cdot)\rangle = 0$, which places k_u in $\operatorname{range} K$. In particular, $KK^+ k_u = k_u$, so the algebra below also applies under the pseudoinverse convention.

Put $A = K + n\lambda I$ and $w = A^{-1}k_u$. For each component j, membership $r_j \in \mathcal{H}_k$ and the reproducing property give

$$r_j(u) - \widehat{r}_{\lambda,j}(u) = \Big\langle r_j,\ k(u, \cdot) - \sum_{i=1}^n w_i\, k(u_i, \cdot)\Big\rangle_{\mathcal{H}_k},$$

because $\widehat{r}_{\lambda,j}(u) = k_u^\top A^{-1} r_j(X) = \sum_i w_i\, r_j(u_i)$ and $r_j(u_i) = \langle r_j, k(u_i, \cdot)\rangle$. Cauchy–Schwarz yields

$$\big|r_j(u) - \widehat{r}_{\lambda,j}(u)\big| \ \le\ \|r_j\|_{\mathcal{H}_k} \Big\|k(u, \cdot) - \sum_i w_i k(u_i, \cdot)\Big\|_{\mathcal{H}_k},$$

and the second factor is independent of j; call it $\rho(u)$. Expanding,

$$\rho(u)^2 = k(u, u) - 2\, w^\top k_u + w^\top K w = k(u, u) - k_u^\top A^{-1}\big(2A - K\big)A^{-1} k_u.$$

Since $2A - K = A + n\lambda I$,

$$\rho(u)^2 = k(u,u) - k_u^\top A^{-1} k_u - n\lambda \, \|A^{-1}k_u\|_2^2 = \widetilde{P}_\lambda(u)^2 \ \le \ P_\lambda(u)^2.$$

Summing the squared componentwise bounds,

$$\left\| r(u) - \widehat{r}_\lambda(u) \right\|_2^2 = \sum_j \left(r_j(u) - \widehat{r}_{\lambda,j}(u) \right)^2 \le \rho(u)^2 \sum_j \|r_j\|_{\mathcal{H}_k}^2 = \widetilde{P}_\lambda(u)^2 \, \|r\|_K^2.$$

For the sharpness, write $g_u = k(u,\cdot) - \sum_i w_i k(u_i,\cdot)$, so that $\|g_u\|_{\mathcal{H}_k} = \rho(u) = \widetilde{P}_\lambda(u)$. If $\widetilde{P}_\lambda(u) = 0$ both sides of the bound vanish for every r. Otherwise take $r_1 = (\|G - m\|_K/\rho(u))\, g_u$ and $r_j = 0$ for $j \ge 2$: this residual has norm $\|G - m\|_K$, and since the estimator is built from its own training values $r_1(X) = (\|G - m\|_K/\rho(u)) g_u(X)$, the first display gives $r_1(u) - \widehat{r}_{\lambda,1}(u) = \langle r_1, g_u \rangle_{\mathcal{H}_k} = \|G - m\|_K \, \rho(u)$, equality in Cauchy–Schwarz. □

The argument applies for every $\lambda \ge 0$. The inequality between $\widetilde{P}_\lambda$ and P_λ compares the two factors at the same ridge value; Theorem 6.10 describes how that value affects the worst-case error.

S8.20 Proof of Theorem 6.10

Write $k_u = k(X,u)$ and let $g_u = k(u,\cdot) - \sum_i (K^{-1}k_u)_i k(u_i,\cdot)$ be the function of the previous proof at $\lambda = 0$. It vanishes on the design, since $g_u(u_j) = k(u,u_j) - k_u^\top K^{-1} K e_j = 0$, and $g_u(u) = \|g_u\|_{\mathcal{H}_k}^2 = k(u,u) - k_u^\top K^{-1} k_u = P_0(u)^2$. If $P_0(u) = 0$ the lower bound is trivial. Otherwise set $r^\pm = \pm(\rho/P_0(u))\, g_u \, e_1$, two residual operators of norm ρ whose training values are both zero, so that any Ψ returns the same vector $\psi = \Psi(0)$ for both; their values at u are $\pm\rho P_0(u) e_1$, hence $\max_\pm \|r^\pm(u) - \psi\|_2 \ge \max_\pm |\pm\rho P_0(u) - \psi_1| \ge \rho P_0(u)$. This is the lower bound. Theorem 6.9 with $\lambda = 0$ bounds the interpolant's error by $\|r\|_K P_0(u) \le \rho P_0(u)$ on the whole ball, so the interpolant attains it, and for $\lambda > 0$ the same theorem together with its sharpness case gives the ridge correction's worst case as exactly $\rho\, \widetilde{P}_\lambda(u)$.

For the identity, put $t = n\lambda$ and $A = K + tI$, which commutes with K. From the proof of Theorem 6.9, $\widetilde{P}_\lambda(u)^2 = k(u,u) - k_u^\top A^{-1}(2A - K)A^{-1} k_u$, while $P_0(u)^2 = k(u,u) - k_u^\top K^{-1} k_u$, so

$$\begin{aligned}
\widetilde{P}_\lambda(u)^2 - P_0(u)^2 &= k_u^\top \left(K^{-1} - 2A^{-1} + A^{-1} K A^{-1} \right) k_u \\
&= k_u^\top A^{-2} K^{-1} \left(A^2 - 2AK + K^2 \right) k_u \\
&= k_u^\top A^{-2} K^{-1} (A - K)^2 k_u,
\end{aligned}$$

and $A - K = tI$. The right-hand side is nonnegative because $A^{-2}K^{-1}$ is positive definite, which gives $P_0 \le \widetilde{P}_\lambda$; the upper inequality $\widetilde{P}_\lambda \le P_\lambda$ was shown above. Expanding k_u in the eigenbasis of K as in the proof of Lemma S1.7, the difference equals $\sum_i (\beta_i^2/\mu_i)\,(t/(\mu_i + t))^2$, each term of which is bounded by β_i^2/μ_i and tends to zero with t, so the difference vanishes as $\lambda \to 0$ by dominated convergence. □

S8.21 Proof of Proposition S3.1

Condition on the fixed fitted predictor, scale function and design. The calibration scores $s_1, \dots, s_m$ and the new score $s^\star$ are exchangeable, and coverage is the event $s^\star \le Q$. If $k = m + 1$ then $Q = +\infty$ and coverage is one. Suppose therefore that $k \le m$.

Assign independent continuous random tie breakers to the $m+1$ scores and let J be the rank of the new score in this augmented ordering. By exchangeability, J is uniform on $\{1,\dots,m+1\}$. The event $J \le k$ implies $s^\star \le s_{(k)}$, including in the presence of ties, so

$$\mathbb{P}(s^\star \le Q) \ge \mathbb{P}(J \le k) = \frac{k}{m+1} \ge 1-\alpha.$$

When the scores are almost surely distinct the two events coincide; their probability equals $k/(m+1) \le 1-\alpha+1/(m+1)$. The same upper bound also includes the already handled case $k = m+1$. Averaging over any independent fitted objects preserves these statements. $\square$

S8.22 Proof of Corollary S5.2

The nonzero eigenvalues of $K/n = XX^\top/n$ coincide with those of the sample covariance matrix $S = X^\top X/n \in \mathbb{R}^{p\times p}$, and zero eigenvalues contribute nothing to $d_{\text{eff}}(\lambda) = \sum_i \lambda_i(K)/(\lambda_i(K)+n\lambda)$, so

$$\frac{d_{\text{eff}}(\lambda)}{n} = \frac{p}{n}\cdot\frac{1}{p}\sum_{j=1}^{p}\frac{\lambda_j(S)}{\lambda_j(S)+\lambda} = \frac{p}{n}\Big(1-\lambda\,\widehat{m}_S(-\lambda)\Big),$$

by the computation of Lemma S5.1 applied to S, with $\widehat{m}_S$ the Stieltjes transform of the empirical spectral distribution of S. By the Marčenko–Pastur theorem (Marčenko and Pastur, 1967), this distribution converges weakly, almost surely, to the law with ratio γ and scale σ^2, and since $x \mapsto (x+\lambda)^{-1}$ is bounded and continuous on $[0,\infty)$, $\widehat{m}_S(-\lambda) \to m(-\lambda)$. It remains to evaluate $m(-\lambda)$. The Stieltjes transform of the Marčenko–Pastur law satisfies the self-consistent equation

$$m(z) = \frac{1}{\sigma^2\big(1-\gamma-\gamma z\,m(z)\big)-z},$$

i.e. $\gamma\sigma^2 z\,m^2 + \big(z-\sigma^2(1-\gamma)\big)m + 1 = 0$. At $z=-\lambda$ the discriminant is $\big(\lambda+\sigma^2(1-\gamma)\big)^2 + 4\gamma\sigma^2\lambda > 0$, and of the two real roots

$$m(-\lambda) = \frac{\sigma^2(1-\gamma)+\lambda \pm \sqrt{\big(\lambda+\sigma^2(1-\gamma)\big)^2+4\gamma\sigma^2\lambda}}{-2\gamma\sigma^2\lambda}$$

only the one with the minus sign in the numerator is positive, as $m(-\lambda) = \int (x+\lambda)^{-1}dF(x) > 0$ requires; rearranging gives the stated expression. $\square$

S8.23 Proof of Lemma S5.1

Diagonalize K with eigenvalues λ_i. Then

$$\begin{aligned}\operatorname{tr}\big(K(K+n\lambda I)^{-1}\big) &= \sum_{i=1}^{n} \frac{\lambda_i}{\lambda_i + n\lambda} \\ &= \sum_{i=1}^{n} \Big(1 - \frac{n\lambda}{\lambda_i + n\lambda}\Big) \\ &= n - \lambda \sum_{i=1}^{n} \frac{1}{\lambda_i/n + \lambda} \\ &= n\big(1 - \lambda\, \widehat{m}(-\lambda)\big),\end{aligned}$$

using $\widehat{m}(-\lambda) = \frac{1}{n}\sum_i (\lambda_i/n + \lambda)^{-1}$. The limit statement follows from the weak convergence of the empirical spectral distribution together with the boundedness and continuity of $x \mapsto (x+\lambda)^{-1}$ on $[0,\infty)$ for fixed $\lambda > 0$. $\square$

S9 Implementation details

The pipeline was developed on a single laptop (one NVIDIA RTX PRO 2000, 8 GB; 16 CPU cores, 64 GB RAM); the structural-mechanics numbers were then recomputed from scratch on CPU-only cloud instances under the seed protocol below, in single precision for network training and double precision for all kernel solves and error computations, and it is those recomputed values that the tables report; the second campaign of Section 4.1 ran on one Intel Xeon E5-2698 v4 host (20 physical cores, 40 hardware threads) under the same protocol. The run records carry no device field, but each network trainer prints the device it selected when it starts, and the ten structural-mechanics pipeline logs hold that line for every launch, restarts included: 54 launches, all `cpu`. The kernel stages are NumPy and SciPy and run on the CPU by construction. The dataset is the distributed `StructuralMechanics` file pair (40000 samples); inputs are reduced to $\mathbb{R}^{41}$ after verifying the broadcast structure exactly. Splits: the training block is samples 1–20000, the test set is samples 20001–40000. High-data protocol: a fixed permutation of the training block reserves 1000 samples for validation, 19000 for fitting. Low-data protocol: the first 1250 samples of the training block, of which the last 250 form the validation split. Model-level selection uses validation.

FNO. Width 64, 14 modes per dimension, 4 spectral layers with pointwise linear skips and GELU, lift from 3 channels (broadcast load, two coordinates), projection $64 \to 128 \to 1$. AdamW, learning rate 1.5×10^{-3} (2×10^{-3} with batch 256), weight decay 10^{-6}, cosine schedule to 10^{-6}, 200 epochs, batch 256. 6.45M parameters.

Transformer (implemented, not in the reported ensemble). Encoder: 41 tokens (load value and 10 sine/cosine pairs of the coordinate), 5 pre-norm blocks, dimension 192, 4 heads, MLP ratio 4. Decoder: grid queries from Fourier features of both coordinates, two cross-attention blocks, pointwise head $192 \to 192 \to 1$; 3.16M parameters. The cross-attention over 1681 grid queries is the most compute-intensive of the means; it is not part of the reported ensemble.

UNet. Three convolutional scales ($41 \to 21 \to 11$ by average pooling, ceil mode) and a bottleneck at 6×6; two 3×3 convolutions with GroupNorm(8) and SiLU per block; widths 48/96/192/384; bilinear upsampling with skip concatenation; 1×1 output head. Input channels as for the FNO. AdamW, learning rate 1.5×10^{-3}, weight decay 10^{-5}, cosine schedule, 200 epochs, batch 256. 4.38M parameters.

MSE variant. The residual MLP trained with mean squared error on standardized targets in place of the metric loss, otherwise identical recipe but for the schedule: the campaign ran it for 120 epochs; it enters the ensemble as a loss-diversity member and is also the strongest plain MLP we obtain.

MLP. Input layer $41 \to 1024$, three residual SiLU blocks of width 1024, output $1024 \to 1681$. AdamW, learning rate 10^{-3}, weight decay 10^{-5}, cosine schedule, 400 epochs, batch 256. 4.91M parameters. A wider variant (width 1536, five residual blocks, 14.5M parameters) is trained under the same recipe.

Refiner. The same residual trunk with input layer $(41 + 1681) \to 1024$: the load is concatenated with the kernel method's stress prediction for that load. Training uses four-fold out-of-fold kernel predictions for the input channel and full-data kernel predictions at evaluation; otherwise identical to the MLP but run for 100 epochs (the trainer's default is 300). 6.64M parameters. Reflection augmentation flips the load and permutes both the kernel channel and the target by the row-reversal index. Evaluation averages $F(u, h(u))$ with $TF(Su, Th(u))$; it does not recompute the kernel prediction at Su. The full-map guarantee and the additional condition on the kernel channel are distinguished in Supplement S1.1.

Common training details. Loss: mean over the batch of $\|\hat{v} - v\|_2 / \|v\|_2$ in original units (predictions denormalized inside the loss; targets standardized by the per-pixel training mean and global standard deviation), except for the MSE variant above. Reflection augmentation with probability 1/2 for neural members; reflection averaging at evaluation, with the refiner's supplied-field convention just described. The kernel-only member has neither operation. Model selection: validation error computed every 10 epochs, best checkpoint kept. The ensemble's diversity comes from the architectures and losses rather than from reseeding; the correlation analysis of Section 4.5 indicates same-architecture reseeds would be more correlated still.

Test-block use. The structural-mechanics test block is the benchmark's fixed public split. It is used by the first-campaign tables, the six-member campaign, the ablations, the hard-case study, the learning curves, the sixty-predictor analysis, the conformal calibration (which draws its 1000 calibration cases from it) and the additional controls. Predictor fitting and model-level selection use the fitting and validation splits. The test-block analyses additionally fit the explicitly labelled hindsight ensemble optimum and choose calibration quantiles on a test subset. The 2000 public OCO-2 test states are in the same position across the ten-seed campaign, the readout control and the ensemble scoreboard.

Training labels and target centering. Only the kernel channel uses foldwise fitted kernel weights. Its predictions use a pooled target mean over all 19000 fitting rows, so held-out fold targets enter that centering statistic. Neural training predictions are generated by the networks fitted on those rows, and the refiner uses its training kernel channel. The correction is therefore fitted to $Y - M_{\mathrm{tr}}$. At query time the mean uses full-data kernel predictions. Section 6.4 gives the explicit $m_{\mathrm{full}}(X) - M_{\mathrm{tr}}$ mismatch term. Supplement S11 measures its propagation with the original mean and kernel fixed, and separately reruns target centering with downstream refiner training and fitting repeated. The two sensitivity experiments are reported separately.

Seed protocol. In the high-data campaign, one seed value enters the stochastic components: the permutation that draws the validation split from the training pool, each member's initialization and batch order, the half-split of the stacking comparison, the subsamples used to tune the kernel stages, and the conformal calibration draw; the pool/test boundary itself never moves. In the low-data protocol the first 1000 fitting and next 250 validation rows remain fixed across seeds. ClimSim is the one exception, and deliberately so: that benchmark fixes no test split of its own, so the seed there also draws the 20000-row test block out of the full set (`campaign/climsim_seeded.py`). Its seed spreads therefore include test-sampling variation, which the structural-mechanics and OCO-2 spreads exclude by construction. These standard deviations therefore describe different sources of variation. Tables report the mean and standard deviation across seeds. What the repository carries is the per-seed result record for every stage of every run — the JSON files the tables and this appendix are computed from, together with the scripts that compute them. The network checkpoints and full prediction/residual arrays are not bundled in the repository and are available from the author on request; the JSON summaries support table and matrix-level checks.

Additional computations. Three computations use the stored campaign arrays or the campaign's own scripts, and each is recorded under `collected/dgx`. First, the conformal bands of Section 6.5: `campaign/uq_conformal_plam.py` computes the P_λ-scaled band at every seed, rebuilding P_λ from each seed's own correction kernel and using the same calibration split as the constant-radius and disagreement-scaled bands, and Section 4.7 reports all three. Second, the frozen-feature readout control for the OCO-2 kernel heads: `campaign/jpl_seeded.py` runs at ten seeds per band with a ridge readout on the same frozen features as an additional row (Table 7). Third, the ClimSim kernel at larger budgets: `campaign/climsim_seeded.py` takes the fitting budget as an argument, and Supplement S7 reports the kernel's own ladder under the same protocol.

End-to-end cost of the kernel stages. The parameter counts of Table 4 are the networks' weights alone. A deployed structural-mechanics pipeline also stores the coefficient matrix of the kernel member and that of the residual correction, each 19000×1681, about 31.9 million numbers apiece (255 MB each in double precision), and evaluates a kernel row against the 19000 training loads for every query. Measured at the campaign's shapes on the campaign host (`campaign/dgx_checks/cost_check.py`, eight threads, synthetic values so that only the arithmetic is timed): the block-filled Gram build takes 27 s and the Cholesky factorization 22 s, the triangular solves with 1681 right-hand sides 11 s, and the peak resident memory of that diagnostic path is 6.0 GiB against 19.2 GiB after the naive full-matrix build. The timing diagnostic demonstrates this memory reduction; the campaign's `correct_stack.py` still forms full-size temporaries. At inference one kernel stage costs 14 ms for a single query and 1.7 ms per query in batches of a thousand. These timings measure one kernel stage. The full six-member pipeline evaluates two such stages plus five neural members with reflection averaging; its total inference latency was not measured by this diagnostic.

What the two campaigns completed. The first structural-mechanics chain was enqueued at ten seeds on four CPU instances. The kernel ridge, the metric-loss MLP, the normalized-MSE MLP and the refiner completed at all ten, forty trained artifacts in total. The campaign's 36-hour per-task limit then landed inside the FNO stage at every seed. The FNO has a best-validation checkpoint at all ten seeds, written by the trainer each time validation improved, and those checkpoints are finalized by running the tail of the trainer on them (evaluate, then save the named artifact), so the FNO of that campaign is a ten-seed member and nothing about it is retrained. Its training curves bound what the early stop cost: the best validation epoch falls between 10 and 70 of the scheduled 200, and at every seed validation was rising again by the last epoch reached, in several cases sharply (seed 3, 0.0489 at its best against 0.0550 at epoch 150; seed 5, 0.0483 against 0.0557). The best observed checkpoint is retained. The UNet, scheduled after the FNO, did not run inside that limit. With the FNO finalized, all ten seeds carry the five-member stack, the residual correction, the second-moment measurement and the conformal calibration; the four-member versions of each are retained and reported so that the FNO's contribution is a measurement rather than an assumption. One instance was killed by the operating system while forming the 19000×19000 Gram matrix through full-size temporaries (the block-filled diagnostic in Section S6.4 measures a lower-memory alternative), and four seeds hit the task limit during stack stages and were completed on a larger instance afterwards.

The second chain ran the same ten seeds on one host with 20 physical cores and 40 hardware threads. It trained the kernel ridge and the three MLP-family members again under the same seeds — their test errors agree with the first campaign's to within 6×10^{-8} in the relative-error unit for the two MLPs and the kernel ridge and 6×10^{-6} for the refiner — and then the FNO and the UNet under a 100-epoch cosine schedule, each to the end of it: nine of the ten FNO runs and all ten UNet runs keep their final checkpoint, the tenth FNO its epoch-79 one. The schedule is normalized to its own budget, so it anneals fully to 10^{-6}; it is a complete shorter schedule, not the first campaign's schedule cut short. The per-pixel stack, the residual correction, the global convex stack, the second-moment measurement and the conformal calibration were then run on the six members at every seed and, for the paired comparisons of Section 4.3, on the same five without the UNet; in that five-member run the half-split rule declined the per-pixel weights at one seed, where the deployed pipeline is the corrected global convex stack, as the rule prescribes. The member cost is recorded in Table S9.2.

The OCO-2 and ClimSim sweeps of Section S7.2 report their own seed counts, which are ten per band and five, three and three by training size respectively. The low-data chain completed at ten seeds. The OCO-2 head pipeline is replicated at ten seeds on each of the three bands — thirty band-seeds at a matched 250-epoch budget, with one further run at 750 epochs to separate budget effects from ordering effects. Each band-seed writes its member predictions and per-sample errors, which is what makes the combination rules, the floors and the scoreboard of Section 5 recomputable without retraining; those arrays stay with the run and are not part of the distributed release. Seven of the thirty band-seeds were run a second time on different hardware, which serves as a reproducibility check: six of the ten rows reproduce to the printed precision across the two runs, including the kernel rows, the flat-metric network, its feature head and the combination that the paper reports, while the three weighted-loss members and their heads move by up to 0.39 points. Seeding fixes the initialization and the data order but not every reduction the training performs, so a seeded rerun of those members should be expected to land nearby rather than exactly; the tables' standard deviations, which are over seeds, are larger than this in every case.

Reconciling the stored arrays. Every member's saved prediction arrays are checked against the metric recorded in its own training record before any ensemble is formed: the array is rescored under the project's metric and the two must agree. The check matters because each seed has its own validation partition, and scoring one seed's predictions against another's split produces errors near 28% that are otherwise easy to mistake for a training failure — and, worse, produces no discrepancy at all on the one seed whose split happens to be the reference. Reconciliation is run at every seed and every member. On the test block, which no seed moves, all forty pairs agree with their recorded reflection-averaged error to between 10^{-14} and 6×10^{-8}, the precision at which the arrays are stored.

Retained residual archives. The residual and selection analyses of Sections 4.5 and 4.3 consume the member residuals $r_m = f_m - y$ rather than the predictions, and the residuals are retained in compressed archives. For each of the ten seeds one compressed NumPy archive, `sm_residuals_s`k`.npz`, holds the test-block residual of each of the five members and of the reported pipeline (per-pixel stack plus kernel correction) as a 20000×1681 array, the validation-split residual of each member as a 1000×1681 array, the target norms $\|y\|$ of every sample on both splits, and the index arrays of both splits; the residuals are stored in IEEE half precision (`float16`), the norms and indices at full width. The archives were written from the reconciled arrays above by `campaign/export_residuals.py`, which also recomputes every member's validation and test error from the half-precision residual and compares it with the stage record the tables are built from: over all 110 arrays the two agree to within 5.4×10^{-8} in the relative-error unit, and the largest residual in any array is 192 against target norms of 3.7×10^3 to 1.7×10^4. The measured rounding effect on these mean errors is far below their seed spread. At 384 MB per seed, 3.84 GB in all, the archives exceed what the code repository can carry. They are not part of the repository and are available from the author on request; the repository keeps the manifest (`runs/residuals_manifest.json`: per array the seed, the split seed, member, split, shape, dtype, the SHA-256 of its bytes, the largest residual and the three errors just described; per archive its size and SHA-256), so a deposited file can be checked byte for byte against the version the paper was computed from. From these files alone, without the dataset, a reader can recompute the first campaign's member errors in Table 4 as the mean of $\|r_m\|/\|y\|$ over samples; the second-moment matrix S, the simplex weights, the predicted and realized risks and the dispersion factor of Proposition 6.1 (`campaign/secmom_seeded5.py` does nothing else with the arrays than form $r_m/\|y\|$ on the validation split and evaluate $\|\sum_m w_m r_m\|/\|y\|$ on test, using the sum-to-one weight identity on the stored residuals); the pairwise residual correlations and the equicorrelated floor; the equal-weight and fitted ensembles, the spatial error maps and the hard-sample comparison of Section 4.5; and the raw conformal scores $\|r\|$ of the pipeline, calibrated on the same seed-drawn subset of the test block. Such recomputation uses the rounded residuals; it is not a bitwise reproduction of the full-precision analysis. The per-pixel affine refit and the kernel correction need the predictions themselves, $f_m = y + r_m$, and the kernel's posterior scale needs the Gram matrix; both follow from the distributed dataset and the released code. The network checkpoints, the training-split predictions, second-campaign member arrays, and OCO-2 and ClimSim arrays remain with their respective runs. The five-member archive does not contain the second campaign's completed FNO or added UNet.

Compute record. The campaign recorded wall clock but not memory; Table S9.1 reports those times. Each member's trainer writes its own elapsed minutes and the epoch of the checkpoint it kept into its run record, prints a cumulative timer every ten epochs, and prints the device it selected; the kernel stage prints the elapsed seconds of every grid cell, one Cholesky factorization and solve of the 19000×19000 system followed by the validation prediction, and the out-of-fold stage the seconds per fold at $n = 14250$. Every task ran six threads, five tasks to an instance, so the figures are wall clock under the campaign's own load rather than the cost of an isolated job, and the spread across seeds is mostly the spread across hosts: the four seeds the larger instance lists as its own in the campaign's task records (2, 4, 8 and 9) are separated from the other six by the kernel's per-cell time without overlap, 20–35 s against 42–113 s, and the table keeps the two classes apart. In this campaign the FNO has no completed record, and its rate is read from the ten-epoch timer prints: the uninterrupted intervals run at 467–625 s per epoch on the standard instances and 348–410 s on the larger one, and the remaining intervals, which reach 10^4 s per epoch, record contention and suspension on the shared hosts rather than the method, which is why the table gives the median of all intervals alongside the minimum. The checkpoints that were kept sit at epochs 10 to 70 and were reached after 1.3–8.0 h of training at the five seeds whose timer ran uninterrupted to that epoch, and after 9.0–16.9 h at the other five, whose timers include such periods. The inference time for the test block was not recorded for any member (the prediction logs carry the score only), and no stage recorded peak memory; the out-of-memory marker at seed 4 is the Gram-matrix failure described above. The harvest, with the per-seed values behind every entry, is `runs/timing_harvest.json`, produced by `campaign/harvest_timing.py` from the run records and logs.

What the members cost. Table S9.2 is the second campaign's cost record at all ten seeds, every member trained to the end of its schedule on one host. Thread counts differ across members because the host was shared with the rest of the campaign. We report thread-minutes, elapsed wall time multiplied by the configured thread count. This records allocated parallelism, not measured CPU utilization, and remains sensitive to contention on the shared host. The FNO uses about 7900 thread-minutes against 54 for the MSE MLP while improving single-member error by about 0.03 percentage points. The UNet uses about 115 times the MSE MLP's thread-minutes and has higher single-member error, although it improves the ensemble. Parameter counts alone do not explain these differences: the FNO has 6.45M parameters, the refiner 6.64M, and the UNet 4.38M. Convolution over the full field and dense prediction from the load vector have different computational costs; the table measures their implementations at the stated schedules, rather than intrinsic costs of the architecture families.

Stacking. Global weights: convex, parametrized by logits, started uniform and optimized by a random hill climb on the validation metric (`stack_correct_seeded.py`: 600 proposals, Gaussian step 0.3 decaying by 0.3% per step, fitted on half of the validation split). The

Table S9.1: Compute record of the structural-mechanics members, as written by the campaign's own trainers and logs (all stages on CPU, six threads per task, five tasks sharing each instance). Network rows: the trainer's elapsed minutes from the first epoch to the final evaluation, as the range over seeds, with the scheduled epochs and the epoch of the kept checkpoint (range over the ten seeds). The FNO row gives seconds per epoch from the ten-epoch timer prints, minimum and median over all intervals (count in parentheses), since no seed completed the schedule. Kernel rows: the whole stage (fifteen grid cells and the refit) and the single factor-and-solve it repeats, as ranges, with the number of timed cells or folds in parentheses. The two columns are the two instance classes of the campaign, with the seed count behind each.

Stage	Epochs: scheduled / kept	Standard instances (6 seeds)	Larger instance (4 seeds)
Metric-loss MLP	400 / 39–79	129–158 min	26–35 min
MSE MLP	120 / 99–119	9.7–11.9 min	6.6–11.1 min
Refiner	100 / 99	9.9–13.1 min	7.9–33.7 min
FNO, per epoch	200 / 10–70	467 s min, 581 s median (88)	348 s min, 437 s median (56)
Kernel ridge, grid and refit	—	17.0–26.0 min	6.1–7.4 min
one factor-and-solve, $n = 19000$	—	42–113 s (90)	20–35 s (60)
out-of-fold fold, $n = 14250$	—	32–66 s (24)	13–22 s (16)

Table S9.2: Cost record of the second campaign on a shared CPU host with 20 physical cores and 40 hardware threads, every member trained to the end of its schedule; ranges over the ten seeds. Wall minutes are the trainer's own elapsed record; thread-minutes is wall clock times the thread count (10 for the convolutional members, 10 to 16 for the MLPs, which is why those ranges are wide); the kernel stage does not write a thread count. Test errors are reflection-averaged where the member has a reflection, as in Table 4. The harvest is `campaign/collected/dgx/runtime_ledger.csv`, produced by `runtime_ledger.py` from the run records.

Member	Parameters	Epochs	Wall (min)	Thread-min	Test error (%)
Kernel ridge, grid and refit	—	—	5.8–10.0	—	5.193–5.200
Metric-loss MLP	4.91M	400	8.8–19.2	132–192	4.809–4.858
MSE MLP	4.91M	120	2.6–5.8	39–58	4.703–4.719
Refiner	6.64M	100	2.6–6.1	41–61	4.721–4.742
FNO	6.45M	100	662–836	6618–8355	4.665–4.697
UNet	4.38M	100	573–689	5734–6892	4.683–4.793

simplex minimizer of the second-moment matrix S (Proposition 6.1) appears only in the pool analyses. Per-pixel weights: affine in the members at each grid point, ridge parameter 10^{-3}, fitted on half the validation split and accepted only if they beat the global weights on the held-out half, then refitted on the full split.

Kernel stages. Matérn-5/2 on standardized inputs. The residual correction searches the scale grid $\{1, 2, 4\}\times$ the median pairwise distance (estimated on 2000 points) and the nugget grid $\{10^{-6}, 10^{-5}, 10^{-3}\}$ (scaled by n); the low-data chain of Table 3 uses $\{0.5, 1, 2, 4\}$ with nuggets $\{10^{-7}, 10^{-5}, 10^{-3}\}$. Both are tuned on validation using an 8000-sample subsample of the training set, refit at the chosen pair on the full training set in double precision.

The pure kernel baseline (Table 2) uses the grid $\{0.5, 0.75, 1, 1.5, 2\}\times$median and nuggets $\{10^{-8}, 10^{-6}, 10^{-4}\}$ directly at $n = 19000$. Feature stage: penultimate activations of the ensemble members (FNO: spatially averaged pre-projection channels, 128; transformer: mean-pooled encoder tokens, 192; MLP: trunk output, 1024), concatenated, standardized, same kernel family and tuning.

Uncertainty. Posterior standard deviation (4) computed from the Cholesky factor of $K + n\lambda I$ by triangular solves. Conformal calibration uses a random 1000-sample subset of the public test block and evaluation uses the disjoint remainder. Model-fitting optimizers use training and validation. The fresh-population guarantee of Proposition S3.1 requires the predictor and scale to be fixed independently of exchangeable calibration and evaluation observations. The exact Beta reference law additionally assumes i.i.d. continuous scores. Reported levels 0.10 and 0.05 have $k \leq m$; for smaller miscoverage with $k = m + 1$, the mathematical quantile is $+\infty$, not the largest finite calibration score.

S10 Numerical verification of the stated results

Numerical checks complement the proofs. The public source snapshot contains synthetic checks and matrix-level checks of the experiments. The checks in this section use those recorded experiments; Supplement S11 reports the paired reruns and prediction reconstruction. The reported numerical macros agree with the released summary records, and independent matrix calculations reproduce the sixty-member empirical RMS floor 4.813611% and recorded optimum 4.876133%.

The second-moment identity is checked on the same arrays used to calculate the quadratic form. The validation dispersion factors 0.938 and 0.938 compare mean and RMS errors on that same sample. In the original five-member bracket record, the normalized value 0.980 ± 0.003 is the held-out risk of fitted global weights, not a simplex minimum. The six-member value 0.972 ± 0.005 is the validation simplex minimum. The two records measure different quantities.

The independence requirements of the capacity bounds fail for validation-adapted candidates in this pipeline; empirical values of their constants do not repair that. Likewise, the fitted residual norms under mixed training labels do not determine the norm of the query-time residual. The correction-label mismatch bound is proved explicitly in the main text. Proposition S2.5 is a finite-design Lipschitz truncation bound with a complete proof. The warped-target scaling experiments are illustrative and do not establish an asymptotic rate theorem.

The reported conformal reference intervals at $m = 1000, N = 19000$ are $[88.0\%, 91.8\%]$ and $[93.5\%, 96.3\%]$ at nominal 90% and 95%. Their Beta–binomial interpretation is stated in Remark S3.2.

The floor theorem, the exchange rate, the sharpened weight bound, the minimax identity and the signed-stack corollary have their own released check, fifteen tests in all, in `campaign/verify_floor_and_certificate.py`. The block floor of Theorem 6.6 is an identity in the exact block-correlated model, and the identity closes to 3×10^{-18} over two thousand random convex weight vectors; on a perturbed pool, and on the released sixty-predictor matrix, twenty thousand random convex mixtures never pass the bound evaluated with the pool's minima, and the mean-entry form of the display returns the uniform mixture's value exactly. The two partial derivatives of Proposition 6.7 agree with central differences to 4×10^{-9} over two thousand interior points, both are positive at the sampled points away from perfect anticorrelation, and along the stated direction the change in the optimal risk is second order in the step. The sharpened bound of Lemma S1.7 holds on fifteen hundred random Matérn designs and evaluation points, the largest ratio to the bound being 0.87, and the cosine-kernel construction attains it to 7×10^{-15}. The identity of Theorem 6.10 closes to 4×10^{-14}, the ordering $P_0 \leq \widetilde{P}_\lambda \leq P_\lambda$ holds on every draw, the residual that the proof of Theorem 6.9 exhibits attains the certified bound with equality to 10^{-14}, and the signed optimum of Corollary 6.5 matches its closed form and never lies above a sampled convex mixture.

These checks cover the displayed algebra at their sampled designs and tolerances.

S11 Implementation sensitivity and prediction reconstruction

These experiments address three implementation choices of the original campaigns: target centering in out-of-fold kernel fits, consistency of correction labels with the query-time mean, and the extent of the OCO-2 kernel search. The run lists and grids were fixed before these experiments started. The $\pm$ values below are sample standard deviations over the stated seeds, not confidence intervals for fresh cases.

S11.1 Paired target-centering reruns

For each mechanics seed, the four kernel folds contain 14250 fitting rows and 4750 held-out rows. The pooled arm subtracts the mean target over all 19000 training rows. The fold-local arm instead subtracts the mean over that fold's fitting rows. Input standardization, kernel scale, nugget 10^{-6}, and fold assignment remain fixed. In particular, "fold-local" here specifies target centering; input standardization still uses the full training block.

The two predictions can be calculated from the same factorization. If $A = K_{FF} + |F|\lambda I$, $k_H = k(X_H, X_F)$, and μ_F, μ_P are the fold and pooled target means, their difference on held-out rows is

$$(\mathbf{1} - k_H A^{-1}\mathbf{1})(\mu_F - \mu_P)^\top.$$

The implementation checks this identity against a separate right-hand-side solve at eight prespecified output coordinates. It also checks that the pooled predictions reproduce the original out-of-fold field before allowing downstream training. The complete per-fold discrepancies are retained in the evidence package.

Each arm retrains the refiner for the same 100 epochs, seed, and eight-thread CPU setting, then repeats global stacking, the held-out decision about per-pixel weights, full-validation refitting, kernel selection, and calibration. The other four neural predictors and the full-training kernel predictor are held fixed. Thus the contrast includes propagation through the downstream fitting steps, rather than merely comparing two kernel fields.

S11.2 Correction-label consistency with a fixed estimator

For each original six-member pipeline, we reconstruct the training-row mean and evaluate its query-time counterpart on those same rows. The selected kernel, its hyperparameters, and the fitted stack weights remain fixed. We then solve for two sets of coefficients: one from the

Table S11.1: Target-centering comparison on the fixed 20000-case mechanics test block, ten paired seeds. The original pipeline is shown as a reference; the two rerun arms use the same training schedule and CPU-thread setting.

Centering convention	Mean relative error (%)
Original retained pipeline	4.5460 ± 0.0029
Pooled centering, paired rerun	4.5459 ± 0.0029
Fold-local centering, paired rerun	4.5459 ± 0.0029
Fold-local minus pooled (percentage points)	0.000011 ± 0.000110

Table S11.2: Calibration after the centering reruns, ten seeds. The same 1000 calibration cases and 19000 evaluation cases are used in each paired arm. Radii are absolute Euclidean norms on the output grid, not percentages or diameters. "Constant" uses scale one; "disagreement" uses the retained ensemble spread. These are measurements on the public test block, subject to the independence qualification in Supplement S3.

		Coverage (%)		Mean radius	
Target	Scale	Pooled	Fold-local	Pooled	Fold-local
90%	Constant	90.40 ± 0.80	90.38 ± 0.77	351.0 ± 4.3	350.9 ± 4.2
90%	Disagreement	90.05 ± 0.40	90.09 ± 0.36	405.8 ± 3.7	406.2 ± 3.9
95%	Constant	95.38 ± 0.54	95.39 ± 0.55	394.3 ± 6.3	394.4 ± 6.4
95%	Disagreement	95.01 ± 0.77	94.98 ± 0.72	466.5 ± 12.6	465.9 ± 11.6

original residual labels, the other from $G(X) - m_{\text{full}}(X)$. The difference of their predictions is exactly the propagated term in (5). The driver checks this coefficient identity and reproduction of the retained validation and test predictions. The aggregation independently checks the reverse-triangle bound on every reported case.

Table S11.3 also reports a two-candidate decision: retain the uncorrected mean or use its inference-consistent correction, according to validation error. This decision does not retune the kernel grid. The separate correction row records the correction whether or not that decision would retain it. The largest propagated relative norm over all test cases and ten seeds is 0.2789%; this maximum is a finite-sample diagnostic, not an upper bound at unseen inputs.

S11.3 A wider OCO-2 kernel search

The grid comparison uses seeds 0, 1, and 2 in each of the three bands. At each band-seed, the three networks are trained for 250 epochs with width 384, then their predictions and features are held fixed for both searches. Both grids use the same training/validation split, the same 6000-row tuning subsample, the same objective for each head, and a refit at the selected hyperparameters on all 18000 training rows.

The original grid has scale multipliers $\{0.5, 1, 2, 4\}$ and nuggets $\{10^{-8}, 10^{-6}, 10^{-4}\}$, for 12 cells per kernel. The expanded grid has multipliers $\{0.25, 0.5, 1, 2, 4, 8, 16\}$ and all eight

Table S11.3: Correction-label sensitivity, ten original mechanics pipelines. No neural training or stack-weight fitting is repeated. The kernel is the one selected in the original run.

Fixed original mean and kernel	Mean relative error (%)
Mean without correction	4.5666 ± 0.0038
Original residual labels	4.5460 ± 0.0029
Inference-consistent residual labels	4.5460 ± 0.0029
Consistent correction or mean, validation-selected	4.5460 ± 0.0029
Consistent minus original (percentage points)	0.000007 ± 0.000029
Mean propagated mismatch, relative norm (%)	0.0148 ± 0.0035

Table S11.4: O2 grid comparison, three paired seeds. The errors are means and sample standard deviations over seeds. Of the 15 selected kernels (five per seed), 13 meet a boundary of the original grid and 5 meet a boundary of the expanded grid. The respective failed-cell counts are 0 and 0.

	Reduced error (%)		Radiance error (%)	
Model	Recorded grid	Expanded grid	Recorded grid	Expanded grid
Source emulator	16.889 ± 0.000	16.889 ± 0.000	0.0448 ± 0.0000	0.0448 ± 0.0000
Raw input	40.529 ± 0.089	40.529 ± 0.089	0.2353 ± 0.0047	0.2353 ± 0.0047
ARD input	30.019 ± 0.035	29.062 ± 0.165	0.2706 ± 0.1030	0.1519 ± 0.0007
Flat network	4.440 ± 0.087	4.440 ± 0.087	0.1893 ± 0.0158	0.1893 ± 0.0158
Flat linear readout	4.774 ± 0.094	4.774 ± 0.094	0.2176 ± 0.0109	0.2176 ± 0.0109
Flat kernel head	4.104 ± 0.074	4.331 ± 0.074	0.0821 ± 0.0092	0.1260 ± 0.0203
Weighted network	49.247 ± 0.008	49.247 ± 0.008	0.0448 ± 0.0036	0.0448 ± 0.0036
Weighted linear readout	46.419 ± 0.328	46.419 ± 0.328	0.0539 ± 0.0009	0.0539 ± 0.0009
Weighted kernel head	21.130 ± 0.281	22.308 ± 0.793	0.0306 ± 0.0010	0.0306 ± 0.0013
Radiance network	59.048 ± 0.406	59.048 ± 0.406	0.0534 ± 0.0047	0.0534 ± 0.0047
Radiance linear readout	44.689 ± 0.909	44.689 ± 0.909	0.0562 ± 0.0043	0.0562 ± 0.0043
Radiance kernel head	20.141 ± 0.361	26.991 ± 0.283	0.0351 ± 0.0008	0.0500 ± 0.0093
Coordinate combination	4.112 ± 0.068	4.329 ± 0.075	0.0282 ± 0.0012	0.0285 ± 0.0012
Combination with linear readouts	4.112 ± 0.068	4.329 ± 0.075	0.0282 ± 0.0012	0.0285 ± 0.0012

powers $10^{-10}, \ldots, 10^{-3}$, for 56 cells. Scales multiply the median distance in the representation being fitted. The isotropic raw-input kernel, ARD-input kernel, and three feature heads all receive these same candidate grids. A Cholesky failure or nonfinite score is recorded as a failed cell; only finite successful validation scores can be selected. The full cell records include selected-boundary flags.

The frozen linear readouts retain their original ridge search, and the source emulator's predictions are unchanged. The coordinate selector is refitted on validation for each grid because its candidate kernel heads change. The tables report every candidate, including the additional selector that also admits the linear readouts. Both errors refer to the retained forty-component representation, as in the main OCO-2 study.

Table S11.5: WCO2 grid comparison, three paired seeds, with the same protocol as Table S11.4. Boundary selections: 9 and 7 out of 15; failed cells: 0 and 0, for the recorded and expanded grids respectively.

	Reduced error (%)		Radiance error (%)	
Model	Recorded grid	Expanded grid	Recorded grid	Expanded grid
Source emulator	24.060 ± 0.000	24.060 ± 0.000	0.0599 ± 0.0000	0.0599 ± 0.0000
Raw input	58.625 ± 0.202	58.625 ± 0.202	0.4431 ± 0.0063	0.4431 ± 0.0063
ARD input	52.454 ± 0.071	53.505 ± 0.073	0.2231 ± 0.0014	0.7523 ± 0.0106
Flat network	16.419 ± 0.070	16.419 ± 0.070	0.2268 ± 0.0049	0.2268 ± 0.0049
Flat linear readout	16.491 ± 0.061	16.491 ± 0.061	0.2502 ± 0.0136	0.2502 ± 0.0136
Flat kernel head	16.270 ± 0.067	16.395 ± 0.073	0.1154 ± 0.0012	0.1770 ± 0.0046
Weighted network	62.501 ± 0.392	62.501 ± 0.392	0.0742 ± 0.0096	0.0742 ± 0.0096
Weighted linear readout	61.294 ± 0.319	61.294 ± 0.319	0.0842 ± 0.0108	0.0842 ± 0.0108
Weighted kernel head	48.826 ± 0.472	50.305 ± 1.924	0.0544 ± 0.0070	0.0626 ± 0.0179
Radiance network	67.505 ± 0.410	67.505 ± 0.410	0.0584 ± 0.0063	0.0584 ± 0.0063
Radiance linear readout	61.206 ± 1.096	61.206 ± 1.096	0.0626 ± 0.0033	0.0626 ± 0.0033
Radiance kernel head	48.247 ± 0.576	50.342 ± 0.397	0.0453 ± 0.0059	0.0558 ± 0.0073
Coordinate combination	16.270 ± 0.068	16.393 ± 0.072	0.0537 ± 0.0069	0.0544 ± 0.0056
Combination with linear readouts	16.270 ± 0.068	16.393 ± 0.072	0.0537 ± 0.0069	0.0544 ± 0.0056

Table S11.6: SCO2 grid comparison, three paired seeds, with the same protocol as Table S11.4. Boundary selections: 15 and 3 out of 15; failed cells: 0 and 0, for the recorded and expanded grids respectively.

	Reduced error (%)		Radiance error (%)	
Model	Recorded grid	Expanded grid	Recorded grid	Expanded grid
Source emulator	16.145 ± 0.000	16.145 ± 0.000	0.1147 ± 0.0000	0.1147 ± 0.0000
Raw input	41.012 ± 0.090	43.708 ± 0.065	0.5229 ± 0.0081	0.5738 ± 0.0054
ARD input	28.717 ± 0.021	28.717 ± 0.021	0.6023 ± 0.0097	0.6023 ± 0.0097
Flat network	8.236 ± 0.021	8.236 ± 0.021	0.2060 ± 0.0209	0.2060 ± 0.0209
Flat linear readout	8.340 ± 0.024	8.340 ± 0.024	0.2436 ± 0.0528	0.2436 ± 0.0528
Flat kernel head	8.092 ± 0.018	8.108 ± 0.025	0.1094 ± 0.0051	0.1142 ± 0.0093
Weighted network	36.028 ± 0.697	36.028 ± 0.697	0.0740 ± 0.0022	0.0740 ± 0.0022
Weighted linear readout	39.605 ± 0.369	39.605 ± 0.369	0.0858 ± 0.0041	0.0858 ± 0.0041
Weighted kernel head	14.908 ± 0.219	15.347 ± 0.194	0.0614 ± 0.0036	0.0609 ± 0.0023
Radiance network	47.386 ± 1.149	47.386 ± 1.149	0.0960 ± 0.0088	0.0960 ± 0.0088
Radiance linear readout	41.112 ± 1.917	41.112 ± 1.917	0.0994 ± 0.0080	0.0994 ± 0.0080
Radiance kernel head	15.506 ± 0.869	19.022 ± 2.830	0.0800 ± 0.0090	0.0904 ± 0.0174
Coordinate combination	8.091 ± 0.018	8.107 ± 0.025	0.0573 ± 0.0045	0.0566 ± 0.0033
Combination with linear readouts	8.091 ± 0.018	8.107 ± 0.025	0.0573 ± 0.0045	0.0566 ± 0.0033

S11.4 Reconstruction from saved prediction fields

The mechanics reconstruction subtracts the saved reference fields in float64, independently recomputes the errors of all sixty retained members, and accumulates the 60×60 evaluation second-moment matrix in blocks of 200 cases. It uses the original fixed 1000/19000 fitting/evaluation split. The largest entrywise difference from the retained matrix is 4.32×10^{-8}.

An independent active-set solve, checked by a convex first-order lower bound, changes the optimal RMS relative error by 1.88×10^{-7} percentage points. The block lower bound and the convex optimum agree with the retained analysis to four decimal places when expressed as percentages. The original records remain available alongside the reconstruction.

The reported mechanics accuracy uses the ordinary Euclidean norm on the 41×41 output grid. As a separate metric check, we also apply tensor-product trapezoidal weights: one in the interior, one half on edges, and one quarter at corners. The common grid-spacing factor cancels in the relative error. For the ten original corrected pipelines, the plain-grid error is $4.5460 \pm 0.0029\%$, while the trapezoidal-grid error is $4.3666 \pm 0.0030\%$. Nested one-dimensional quadrature independently reproduces the weighted calculation. This check does not refit a predictor or rescore the quoted external comparators; the main comparison continues to use the stated plain-grid metric.

S11.5 Evidence and recomputation

The evidence archive contains executed source files, input hashes, stage receipts, per-case error arrays, calibration scores and partitions, and complete kernel-grid records. The table generator first checks the archive's exact file set and hashes, the completed run list, case counts and ordering, paired signs, calibration quantiles, and validation winners. It recomputes reported means from the per-case arrays by a separate summation path. For each of the twenty completed centering arms, a second check rebuilds the relative and absolute errors from the saved fields. It also recomputes the six-member disagreement scale using the identity

$$\frac{1}{M}\sum_{a=1}^{M}(x_a - \bar{x})^2 = \frac{1}{M^2}\sum_{a<b}(x_a - x_b)^2$$

at every output coordinate, followed by the same coordinate average of the standard deviations. This checks the scale through pairwise differences rather than subtraction of the ensemble mean. The OCO-2 prediction check rebuilds all fourteen predictors' scores under both grids in all nine band–seed combinations. It uses a spectral Gram quadratic form, cross-checks selected cases by direct full-spectrum summation, and reconstructs the coordinate selectors from validation predictions. Checkpoints and large prediction fields remain in the retained run directories. Scripts for collection, verification, and table generation are under `campaign/` in the code release.

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